Hepatic Failure/GI/Endocrine Emergencies

Stephanie N. Bass, Pharm.D., FCCP, BCCCP, BCPS

Cleveland Clinic

Cleveland, Ohio

Mollie G. Lumpkin, Pharm.D., BCCCP, BCPS

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Learning Objectives

1. Develop a treatment strategy to help manage and reduce the complications associated with acute liver failure (ALF).
2. Construct a plan for pharmacologic, nutritional, and surgical management of acute pancreatitis based on the severity of an episode.
3. Identify risk factors and treatment options for gastrointestinal fistulas, postoperative ileus, and postoperative nausea and vomiting.
4. Design a treatment plan for patients who present with an acute upper gastrointestinal bleed.
5. Differentiate between the main endocrine emergencies in the intensive care and their appropriate treatment regimens.

Abbreviations in This Chapter

ALF	Acute liver failure
AP	Acute pancreatitis
BG	Blood glucose
CIRCI	Critical illness–related corticosteroid insufficiency
CPP	Cerebral perfusion pressure
CT	Computed tomography
DILI	Drug-induced liver injury
DKA	Diabetic ketoacidosis
ED	Emergency department
ERCP	Endoscopic retrograde cholangiopancreatography
GI	Gastrointestinal
HHS	Hyperosmolar hyperglycemic state
ICP	Intracranial pressure
ICU	Intensive care unit
INR	International normalized ratio
MRI	Magnetic resonance imaging
NAI-ALF	Non–acetaminophen-induced acute liver failure
NG	Nasogastric
NJ	Nasojejunal
NSAID	Nonsteroidal anti-inflammatory drug
POI	Postoperative ileus
PONV	Postoperative nausea and vomiting
PPI	Proton pump inhibitor
SIRS	Systemic inflammatory response syndrome
T_3	Triiodothyronine
T_4	Thyroxine
TIPS	Transjugular intrahepatic portosystemic shunt
TPN	Total parenteral nutrition
TSH	Thyroid-stimulating hormone
UGIB	Upper gastrointestinal bleeding

Self-Assessment Questions

Answers and explanations to these questions can be found at the end of this chapter.

Questions 1 and 2 pertain to the following case.

A 25-year-old woman is brought to the emergency department (ED) after a suspected overdose of acetaminophen. The time of ingestion is unknown. On presentation, her acetaminophen concentration is undetectable, but her alanine aminotransferase (ALT) and aspartate aminotransferase (AST) concentrations are 3500 IU/L and 2500 IU/L, respectively. The patient is markedly confused with incoherent speech, but arousable. Other pertinent laboratory values include bilirubin 3.0 mg/dL and alkaline phosphatase 500 IU/L. White blood cell count is 12×10^3 cells/mm^3, platelet count is 90,000/mm^3, and international normalized ratio (INR) is 2.6.

1. Which option best represents the two signs or symptoms that would qualify this patient for a diagnosis of acetaminophen-induced acute liver failure (ALF)?
   A. Jaundice and encephalopathy.
   B. Thrombocytopenia and encephalopathy.
   C. Coagulopathy and encephalopathy.
   D. Leukocytosis and encephalopathy.

2. Which is the most appropriate treatment for her suspected acetaminophen-induced ALF?
   A. Give intravenous acetylcysteine 21-hour regimen, continuing if necessary until signs and symptoms of ALF have resolved.
   B. Acetylcysteine therapy is not indicated at this time because her acetaminophen concentration is undetectable.
   C. Give oral acetylcysteine 72-hour regimen.
   D. Oral acetylcysteine or intravenous acetylcysteine may be used because the two routes are similarly efficacious.

3. A 46-year-old man presents with alcohol-induced severe acute pancreatitis (AP). Pertinent medical history includes alcoholic cirrhosis and several admissions for aspiration pneumonia secondary to hepatic encephalopathy. He is given intravenous fluids for initial volume resuscitation with lactated Ringer solution. Which action would be best for the patient's nutrition?
   A. Give nothing by mouth (i.e., kept NPO) to rest the pancreas until AP is resolved.
   B. Initiate total parenteral nutrition (TPN).
   C. Give enteral feeding by the nasogastric (NG) route.
   D. Give enteral feeding by the nasojejunal (NJ) route.

4. A 37-year-old woman presents after a Roux-en-Y gastric bypass for morbid obesity. Her postoperative course was complicated by the formation of an enterocutaneous fistula, fevers, and leukocytosis. She was initiated on broad-spectrum antibiotics, and a wound vacuum-assisted closure was placed on her fistula site to help with drainage and healing. Her fistula output was 570 mL/day yesterday, and today, it was 250 mL/day. Which statement is most accurate regarding her fistula output between the two recordings?
   A. Her output would be defined as a low output on both days.
   B. Her output would be defined as a high output that has converted to a low output.
   C. Her output would be defined as a high output that has converted to a moderate output.
   D. Her output would be considered moderate on both days.

5. A 68-year-old man presents for a large bowel resection. Given the high incidence of postoperative ileus (POI) with this procedure, he is initiated on alvimopan before surgery. To reduce the cardiovascular risk associated with alvimopan, the U.S. Food and Drug Administration (FDA) placed a restriction on alvimopan use that has been implemented through the EASE (ENTEREG Access Support & Education) program. Which best represents that restriction?
   A. Start continuous electrocardiogram (ECG) monitoring on initiation to monitor for corrected QT (QTc) prolongation.
   B. Use is restricted to short term with a limit of 15 doses.
   C. Decrease alvimopan dose to 6 mg twice daily.
   D. Use is contraindicated in patients with previous myocardial infarction.

6. A 40-year-old woman presents for elective abdominal surgery to remove a malignancy from her liver. Given the extensiveness of the surgery, the expected use of volatile anesthetics, and the use of perioperative opioids, a multimodal plan is developed to avoid postoperative nausea and vomiting (PONV). Which best describes the timing of prophylactic administration with respect to surgery?
   A. Dexamethasone 4 mg intravenously given at the end of surgery.
   B. Ondansetron 4 mg intravenously given when inducing anesthesia.
   C. Neurokinin-1 receptor antagonist (aprepitant 40–125 mg) given at the end of surgery.
   D. Droperidol 0.625–1.25 mg intravenously given at the end of surgery.

7. A 69-year-old woman presents to the surgical intensive care unit (ICU) at your institution with upper gastrointestinal bleeding (UGIB) caused by a gastric ulcer. She has lost a significant amount of blood because of the bleed and currently requires blood transfusions. As part of her diagnostic workup to determine the etiology of her ulcer, she tests positive for a *Helicobacter pylori* infection. Which best reflects an inappropriate treatment option for her?
   A. Octreotide 50 mcg bolus, followed by 50 mcg/hour for 72 hours.
   B. Treatment with a proton pump inhibitor (PPI)/antibiotic combination for 14 days.
   C. A therapeutic endoscopy within 24 hours.
   D. Blood transfusions to maintain hemoglobin greater than 7 g/dL.

8. Which set of laboratory abnormalities best reflects those that patients in thyroid storm typically present with?
   A. High thyroid-stimulating hormone (TSH), triiodothyronine (T_3), and thyroxine (T_4) concentrations.
   B. Low TSH, high T_3, and high T_4 concentrations.
   C. Low TSH, high T_3, and low T_4 concentrations.
   D. Low TSH, low T_3, and low T_4 concentrations.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 1: Critical Care
   a. Task A: 1, 2, 4
2. Domain 2: Therapeutic and Patient Management
   a. Task A: 1, 4, 6
   b. Task B: 4, 5, 6, 9, 10, 11, 13, 15
   c. Task C: 1

I. ACUTE LIVER FAILURE

A. Epidemiology
1. Incidence of ALF is less than 10 cases per 1 million individuals per year (2000 cases per year in the United States), though morbidity and mortality are exceedingly high with ALF. Multiorgan failure and death occur in as many as 50% of patients.
2. ALF accounts for less than 10% of liver transplants annually in the United States.
3. ALF can occur in any age and demographic group, though the etiology of ALF is sometimes geographically dependent. Viral hepatitis cases are more common in developing countries, whereas toxin-related cases (e.g., those related to acetaminophen) occur more often in developed countries.
4. The most common causes in the United States are drug induced, viral, autoimmune, and shock. Drug-induced causes, primarily acetaminophen, account for about 50% of the ALF cases in the United States. About 15% have no identifiable cause.
5. The mortality rate has decreased significantly during the past few decades because of advances in medical care and early consideration for liver transplantation. Survival rates for ALF now exceed 65%, whereas before early transplantation, survival rates were less than 15%.

B. Definitions
1. ALF is defined by the U.S. Acute Liver Failure Study Group (ALFSG) and the American Association for the Study of Liver Diseases (AASLD) as evidence of liver-induced coagulopathy, defined as an INR of 1.5 or greater, and any degree of mental alteration (encephalopathy) in a patient without preexisting liver failure and with an illness duration less than 26 weeks.
    a. ALF is in contrast to ACLF (acute-on-chronic liver failure), which is a syndrome defined on the basis of acute decompensation of chronic liver disease and presence of multiorgan failure.
2. ALF may be further differentiated according to the time to encephalopathy after the onset of jaundice. These intervals typically provide clues regarding the cause of the ALF; however, the differentiation itself generally has no prognostic implications distinct from the causes themselves.
    a. Hyperacute ALF: Encephalopathy occurs less than 7 days after the onset of jaundice. This subclass of ALF is typically caused by acetaminophen toxicity or ischemic hepatitis and is associated with higher rates of transplant-free survival defined as survival free of liver-related death or liver transplantation. Patients tend to have high-degree encephalopathy at presentation and a higher incidence of cerebral edema, albeit a better prognosis overall.
    b. Acute ALF: Encephalopathy occurs 7–21 days after the onset of jaundice. Common causes of acute ALF include viral hepatitis. These patients have a high incidence of cerebral edema, but unlike in hyperacute ALF, lower rates of transplant-free survival.
    c. Subacute ALF: Encephalopathy occurs more than 21 days and less than 26 weeks after the onset of jaundice. Typical causes are drug induced or indeterminate. This subclass is associated with lower transplant-free survival, though patients have less marked coagulopathy and encephalopathy at presentation and may more often develop renal failure and portal hypertension.

C. Diagnosis and Disposition
1. An unexplained elevated INR in a patient presenting with encephalopathy requires further evaluation for ALF because the combination of these two symptoms is very specific to ALF.
2. In ALF, certain markers of chronic liver disease (e.g., jaundice, ascites, right upper quadrant pain, portal hypertension) may not be present.
3. Additional physical examination, laboratory analysis, and imaging necessary for the diagnosis and workup of ALF are shown in Table 1.

4. Patients with any degree of encephalopathy should be transferred to an ICU, ideally with contact to a transplant center, because rapid progression can occur.

Table 1. Diagnostic Approach to a Patient with Suspected ALF

Diagnostic Work-up	Procedure
History	Assess timing of symptom onset
	Assess for exposure history to viruses, drugs (e.g., acetaminophen), or toxins (e.g., mushrooms)
	Assess substance abuse history
Physical examination	Normal signs of chronic liver disease (e.g., ascites, jaundice, right upper quadrant pain) may not be present
	Assess encephalopathy grade
Laboratory analysis	Basic metabolic panel, CBC, liver function tests, coagulation tests, arterial blood gas, acetaminophen concentrations, ammonia, toxicology screen, blood typing
	Viral serologies
	Liver biopsy useful for determining autoimmune hepatitis or ALF associated with HSV
Imaging	Hepatic imaging studies (computed tomography [CT], ultrasonography) may be used to detect a thrombus of the hepatic vein

ALF = acute liver failure; CBC = complete blood cell count; HSV = herpes simplex virus.

Adapted from: Lee WM. Acute liver failure. Semin Respir Crit Care Med 2012;33:36-45.

Table 2. Causes of ALF in the United States

Cause of ALF	Proportion of Cases
Acetaminophen	46%
Indeterminate	12%
Non-acetaminophen drug induced	11%
Hepatitis B	7%
Autoimmune	7%
Ischemic	7%
Hepatitis A	1.5%
Wilson disease	1.2%
Pregnancy	1%
Budd-Chiari	0.7%
Other	7%

ALF = acute liver failure.

Adapted from: Stravitz RT, Lee WM. Acute liver failure. Lancet 2019;394:869-81.

D. Causes (Table 2)
   1. Acetaminophen
      a. Acetaminophen overdose is responsible for almost 50% of ALF cases in the United States. Acetaminophen overdose is considered the primary cause of ALF in the United States and Europe, and it was responsible for 70,000 health care encounters and 300 deaths in the United States in 2005.

b. Unlike most other types of drug-induced liver injury (DILI), acetaminophen-induced liver failure is dose-dependent and predictable, and it is typically associated with doses above 10 g/day (or more than 200 mg/kg/day) in 24 hours or more than 6 g/day (around 150 mg/kg/day) for 48 hours in adults.
c. Rates of ALF caused by acetaminophen have increased during the previous 2 decades.
d. If not treated in the early stages (i.e., before the development of encephalopathy), the mortality rate is around 20%–40%.
e. Acetaminophen-induced ALF typically presents as hyperacute liver failure and is defined by four stages of progression:
    i. Preclinical: Occurs within the first 24 hours of ingestion. Typically associated with minimal or no signs or symptoms of hepatotoxicity
    ii. Injury: Occurs 24–72 hours after ingestion. Associated with marked elevation in liver transaminases
    iii. Failure: Occurs 72–96 hours after ingestion. Associated with peak liver injury including encephalopathy, coagulopathies, and jaundice
    iv. Recovery: Occurs 1 week after ingestion if patient survives through failure stage. Recovery may be prolonged in critically ill patients who present with fulminant liver failure.
f. Additional information on background, pathophysiology, and treatment of acetaminophen overdose can be found in the Toxicology review chapter.

2. Non–acetaminophen-induced acute liver failure (NAI-ALF)
    a. DILI
        i. The incidence of DILI caused by drugs other than acetaminophen is rare, at a rate of approximately 11% of ALF cases per year.
        ii. Unlike acetaminophen-induced liver failure, DILI is rarely the result of dose-related toxicity, and most cases are idiosyncratic.
        iii. DILI typically presents as a subacute ALF, with most cases occurring within the first 6 months after drug initiation. However, some drugs (e.g., nitrofurantoin, minocycline, statins) have the potential to cause DILI 6 months or more after initiation.
        iv. Transplant-free survival is low for these patients (about 30%), and most patients will require transplantation.
        v. DILI is ultimately a diagnosis of exclusion. The American College of Gastroenterology (ACG) guidelines for the management of DILI recommend a specific workup for viral hepatitis, autoimmune hepatitis, Wilson disease, and Budd-Chiari syndrome before diagnosis of DILI.
        vi. To identify potential culprit medications, a detailed patient medication history should be obtained, including herbal medications. Classes of drugs commonly associated with DILI include antibiotics, nonsteroidal anti-inflammatory drugs (NSAIDs), and anticonvulsants, which together account for almost two-thirds of those attributable to DILI.
        vii. Scoring systems such as the RUCAM (Roussel Uclaf Causality Assessment Method) have been developed to assess the causality attribution for suspected DILI. These scoring systems give points on the basis of timing of exposure and liver function tests, risk factors for DILI, competing medications and diagnoses, and rechallenge information. Higher scores indicate a higher likelihood of drug cause.

viii. The pattern of liver injury may also help identify drug causes. Hepatocellular injury (characterized by a marked elevation in transaminases with marginal increase in bilirubin and alkaline phosphatase) is associated with certain drug causes, whereas cholestatic injury (characterized by a marked elevation in bilirubin and alkaline phosphatase with only minimal increases in transaminases) is associated with other drug causes of DILI. Other drug causes of DILI exhibit a mixed pattern of liver injury in which both hepatocelluar injury and cholestatic injury will be present. See Appendix 1 for a list of medications linked to DILI and associated pattern of liver injury.

b. Viral
   i. Infection with hepatitis A, B, or E is the primary etiology of ALF in the developing world and has become a relatively infrequent cause of ALF in the United States (about 12%).
   ii. Hepatitis A
      (a) Accounts for about 2% of ALF cases in the United States
      (b) Transmission is usually by the fecal-oral route.
   iii. Hepatitis B
      (a) Rates of hepatitis B–induced ALF have fallen significantly in the past few decades; however, hepatitis B is still the cause of about 7% of ALF cases per year.
      (b) ALF secondary to hepatitis B is often caused by reactivation of chronic or inactive hepatitis B during times of immunosuppression (e.g., chemotherapy and high dose steroids). To prevent reactivation, patients who are positive for HBsAg (hepatitis B surface antigen) and who are to begin immunosuppressive regimens require antiviral prophylaxis with a nucleos(t)ide analog, typically entecavir, lamivudine, or tenofovir. Treatment, which is usually initiated before immunosuppression, continues during immunosuppressive therapy and for 6 months thereafter.
   iv. Hepatitis E
      (a) Rare cause of ALF in the United States; however, hepatitis E–induced ALF is a significant cause of ALF in countries where it is endemic, such as Russia, Pakistan, Mexico, and India.
      (b) Hepatitis E–induced ALF tends to be more severe in pregnant women, and hepatitis E can be transmitted to neonates during acute infections in pregnant women.
   v. Herpes simplex virus (HSV)
      (a) HSV is rarely a cause of ALF; however, cases have been reported, particularly in immunocompromised and pregnant patients (usually in the third trimester).
   vi. Viral hepatitis–induced ALF generally presents as acute or subacute liver failure with an onset of symptoms greater than 1 week after onset of jaundice.
   vii. Globally, mortality rates are greater than 50% because of ALF from hepatitis A and E in the developing world; however, mass vaccination and better public health standards have helped reduce the incidence of viral infections in the developed world.

c. Acute ischemic injury
   i. Often called "shock liver," acute ischemic injury may lead to ALF after cardiac arrest, any period of significant hypovolemia or hypotension, or during severe congestive heart failure.
   ii. Documented hypotension is not always evident with acute ischemic injury. Drug-induced hypotension or hypoperfusion may also cause acute ischemic injury, such as with cocaine and methamphetamine.
   iii. Typical laboratory presentation includes markedly elevated aminotransferase concentrations. In addition, simultaneous renal dysfunction and other markers of hypoperfusion may be present.
   iv. Acute ischemic injury is classified as a hyperacute ALF, which generally resolves with resolution of the circulatory problem.

d. Mushroom poisoning
   i. Typically caused by the *Amanita phalloides* spp. of mushrooms
   ii. Mushroom poisoning is classified as a hyperacute ALF with an onset of symptoms within 24 hours after ingestion. In addition to hepatotoxicity, patients with mushroom poisoning present with severe gastrointestinal (GI) symptoms including nausea, vomiting, diarrhea, and abdominal cramping.

e. Wilson disease
   i. Rare cause of ALF; implicated in approximately 1% of cases per year, mostly affecting young people
   ii. Wilson disease is a rare disorder that causes accumulation of copper in the body. Diagnosis is characterized by an abrupt onset of Coombs-negative hemolytic anemia with serum bilirubin greater than 20 mg/dL. Other diagnostic laboratory findings include low serum ceruloplasmin, presence of Kayser-Fleischer rings, and high urinary and hepatic copper concentrations.
   iii. Patients who present with fulminant liver failure secondary to Wilson disease have exceedingly high mortality rates without transplantation.

f. Autoimmune hepatitis
   i. Autoimmune hepatitis is considered a chronic inflammatory disease; however, patients can still be considered to have ALF if they had rapid deterioration of symptoms.
   ii. About 20% of patients with stable disease will have ALF, which is typically instigated by an environmental trigger.

g. Budd-Chiari syndrome
   i. Budd-Chiari syndrome is caused by an acute hepatic vein thrombosis. Presenting symptoms include abdominal pain, ascites, and frank hepatomegaly. Patients who initially present with ALF require anticoagulation with heparin as initial therapy; in those who do not respond anticoagulation, transjugular intrahepatic portosystemic shunt (TIPS) may be considered.
   ii. In patients with ALF who do not respond to medical or therapeutic interventions, transplantation may be required.

h. Acute fatty liver of pregnancy/hemolysis, elevated liver enzymes, low platelets (HELLP)
   i. Toward the end of pregnancy, a small percentage of women will develop rapidly progressive hepatic failure that is generally associated with three hallmark signs: jaundice, coagulopathy, and low platelets, also known as HELLP. Features of pre-eclampsia such as hypertension and proteinuria are common with HELLP.
   ii. HELLP is associated with increased mortality for both the fetus and the mother.
   iii. Requires emergency delivery of fetus, after which symptoms should resolve. If ALF fails to improve after delivery of the fetus, liver transplant evaluation is necessary.

E. Complications
   1. ALF affects almost every organ system in the body:
      a. Neurologic
         i. Cerebral edema and elevated intracranial pressures (ICPs) are the most serious complications of ALF. Uncontrolled edema and elevated ICPs can lead to uncal herniation and are usually fatal. Cerebral edema may also lead to tissue hypoxia, which may result in long-term neurologic deficits.
            (a) Elevated ICPs can be caused by several factors, but osmotic shifts in the brain, inflammation, and neurotoxins are thought to be the primary causes.

(b) Because ammonia is converted to osmotically active glutamine, concentrations have been correlated with both encephalopathy and cerebral edema. Ammonia concentrations greater than 200 mcg/dL are associated with cerebral herniation, whereas concentrations less than 75 mcg/dL are rarely associated with hepatic encephalopathy. In ALF, either because of impaired hepatocyte activity or the abnormal shunting of venous flow away from the liver, normal mechanisms to detoxify and clear ammonia are no longer effective.

ii. Encephalopathy is considered an indicator for the clinical presentation of cerebral edema. Encephalopathy can be difficult to identify and may present initially as agitation and confusion; however, it may progress rapidly to unresponsiveness. Table 3 gives useful guidelines for measuring the severity of encephalopathy.

iii. Occurrence of cerebral edema and elevated ICPs is generally related to the severity of hepatic encephalopathy. Patients with grade I and grade II encephalopathy rarely have cerebral edema, whereas cerebral edema is present in about 25%–35% of patients with grade III encephalopathy and about 65%–75% of patients with grade IV encephalopathy.

Table 3. Grades of Encephalopathy

Grade of Encephalopathy	Signs and Symptoms
Grade I	Changes in behavior with minimal change in level of consciousness
Grade II	Gross disorientation Drowsiness Possibly asterixis Inappropriate behavior
Grade III	Marked confusion Incoherent speech Sleeping most of the time but arousable to vocal stimuli
Grade IV	Comatose Unresponsive to pain, decorticate or decerebrate posturing

Adapted from: Conn HO, Leevy CM, Vlehcevic ZR, et al. Comparison of lactulose and neomycin in the treatment of chronic portal-systemic encephalopathy. A double blind randomized trial. Gastroenterology 1977;72:573-83.

b. Cardiovascular

i. The primary hemodynamic concern in ALF is low systemic vascular resistance, similar to cirrhosis. Additional hemodynamic effects in ALF include hyperdynamic circulation with high cardiac output and decreased effective circulating volume.

ii. Most patients are severely volume depleted on admission because of poor nutritional status and third spacing into the extravascular space and will initially require aggressive fluid resuscitation.

iii. In patients with an elevated ICP, maintaining adequate perfusion is essential to preserve adequate oxygen delivery to the brain. A common goal is to maintain a cerebral perfusion pressure (CPP) of at least 60 mm Hg. Higher mean arterial pressure (MAP) targets may be needed in these patients in order to maintain CPP goals. CPP = MAP − ICP.

c. Coagulopathy

i. ALF is defined by the presence of an elevated INR caused by decreased production, together with increased consumption, of coagulation factors.

ii. Consumption of platelets is also seen, and thrombocytopenia (150,000/mm^3 or less) is common.

iii. Despite these changes in coagulation parameters, thromboelastography studies of patients with ALF have shown that overall hemostasis in patients with ALF is maintained by compensatory mechanisms, even in patients with elevated INR values, potentially because of a reduction in hepatic synthesis of natural anticoagulants.

iv. Spontaneous bleeding in patients with ALF, though uncommon, is capillary-type bleeding and usually results from mucosal bleeding in the stomach, lungs, or genitourinary system. Unlike chronic liver failure, bleeding from esophageal varices generally does not occur.
v. Clinically significant bleeding that requires blood transfusions is rare in ALF.

d. Renal
i. Acute kidney injury with ALF is generally classified as either prerenal injury or acute tubular necrosis.
(a) Prerenal azotemia typically is caused by vasodilatation owing to portal hypertension and is worsened by systemic hypoperfusion, similar to hepatorenal syndrome in patients with cirrhosis.
(b) Acute tubular necrosis typically occurs secondary to drugs or toxins, such as acetaminophen or *Amanita* poisoning.

e. Infection
i. Patients with ALF are at high risk of infection because of prolonged ICU stays, the presence of indwelling foleys and central venous catheters in addition to intrinsic monocyte and neutrophil dysfunction. Infections are of particular concern because they may delay transplantation or be problematic during the postoperative period.
ii. The most common infections in ALF are pneumonia, followed by urinary tract infections and bloodstream infections. The most commonly isolated organisms are gram-positive cocci (e.g., *Staphylococcus*, *Streptococcus*) and enteric gram-negative bacilli. Fungal infections, particularly those caused by *Candida*, occur in about one-third of patients with ALF.

f. Metabolic abnormalities
i. Lack of effective glycogenolysis and gluconeogenesis, caused by impaired hepatocyte function, places patients at high risk of hypoglycemia.
ii. Symptoms of hypoglycemia can often be difficult to identify in patients with severe encephalopathy, whereas profound hypoglycemia can worsen an already altered mental state.
iii. Hyponatremia is also common in patients with ALF, secondary to alterations in antidiuretic hormone from tissue hypoperfusion and renal dysfunction. Avoidance of hyponatremia is particularly important in patients with high-grade encephalopathy.

2. Therapy must be multimodal to support each of the organ systems affected by ALF.

F. Management of ALF
1. Antidotes
a. Acetaminophen-induced ALF (see Table 4)
i. Although most effective if given within the first hour, GI decontamination with activated charcoal may be of benefit for up to 4 hours after ingestion and does not reduce the effect of acetylcysteine.
ii. Administration of acetylcysteine is recommended in all ALF cases in which acetaminophen is suspected as a cause.
(a) Acetylcysteine can be given either orally or intravenously. Studies have shown similar outcomes between the two routes; however, in those studies, the main efficacy outcome of interest was development of hepatotoxicity. In patients already presenting with hepatotoxicity (as in ALF), intravenous acetylcysteine is recommended. The U.S. Acute Liver Failure Study Group recommends intravenous therapy for any of the following:
(1) Greater than grade I encephalopathy
(2) Hypotension
(3) If oral therapy cannot be tolerated (e.g., vomiting, compromised airway, ileus)

(b) Many poison centers may extend therapy beyond the recommended course if there is a detectable acetaminophen concentration or if ALT concentrations continue to remain elevated at the end of therapy, especially if therapy was initiated more than 8 hours after ingestion and baseline acetaminophen concentrations were greater than 300 mcg/mL when extrapolated on the Rumack–Matthew nomogram.

(c) Recent studies have evaluated alternative dosing strategies for patients with massive overdoses, including doubling the last bag (i.e., providing 200 mg/kg over 16 hours). Massive overdoses are considered ingestions with initial acetaminophen concentrations greater than 300 mcg/mL when extrapolated on the Rumack–Matthew nomogram, or a single ingestion greater than 30 g; doubling the last bag has been associated with reduced hepatotoxicity. However, these strategies must be further evaluated.

(d) Therapy may continue until the signs and symptoms of encephalopathy or coagulopathy resolve or until the patient receives a liver transplant. The 2023 ACG ALF guidelines note that duration of NAC treatment for APAP-induced ALF should be "individualized based on patient's clinical condition and laboratory values." (Am J Gastroenterol 2023:118:1128-53)

(e) Additional information on treatment of acetaminophen-induced ALF can be found in the Toxicology chapter.

Table 4. Acetylcysteine for Acetaminophen-Induced ALF

Route	Dose
Oral	Loading dose: 140 mg/kg × 1 dose Maintenance dose: 70 mg/kg every 4 hr × 17 doses (72 hr total)
IV	150 mg/kg (max 15 g) over 60 min, followed by 50 mg/kg (max 5 g) over 4 hr, followed by 100 mg/kg (max 10 g) over 16 hr (21 hr total)

IV = intravenous.

b. NAI-ALF

i. Acetylcysteine may improve oxidative stress in NAI-ALF by acting as a free radical scavenger. In addition, acetylcysteine may improve both hepatic and systemic perfusion through its vasodilatory effects.

ii. A multicenter randomized trial compared intravenous acetylcysteine with placebo for 72 hours (see Table 5) for treatment of NAI-ALF. Randomized patients were stratified according to coma grade, with most patients having a low-grade encephalopathy. There was no difference in the primary outcome of overall survival at 3 weeks between acetylcysteine and placebo; however, the transplant-free survival rate significantly increased with the use of acetylcysteine (40% vs. 27%, p=0.04).

(a) The increase in transplant-free survival with acetylcysteine was mainly confined to the subgroup of patients with encephalopathy grade I and grade II (52% vs. 30% with placebo, p=0.01).

(b) When outcomes were compared on the basis of each etiology of NAI-ALF, patients with DILI and hepatitis B virus had more improvement in overall survival and transplant-free survival from acetylcysteine compared with placebo than with other causes.

iii. Intravenous acetylcysteine has also been studied for the treatment of NAI-ALF as a loading dose followed by a continuous infusion of 150 mg/kg over 24 hours until resolution of coagulopathy (INR less than 1.3). This strategy increased transplant-free survival compared with a historical cohort (96.4% vs. 23.3%, p<0.01); however, the strategy needs to be further evaluated.

iv. Oral acetylcysteine has also been evaluated for the treatment of NAI-ALF in a prospective non-interventional study in which it, compared with a historical cohort, was associated with reduced mortality. In this study, oral acetylcysteine was given as a 140-mg/kg loading dose, followed by 70 mg/kg every 4 hours for an additional 17 doses. However, at this time, only intravenous acetylcysteine has been studied in a randomized controlled trial for NAI-ALF.

Table 5. Acetylcysteine for NAI-ALF

Route	Dose
Intravenous	150 mg/kg over 1 hr, followed by 12.5 mg/kg/hr for 4 hr, followed by 6.25 mg/kg/hr for 67 hr

Patient Case

1. A 60-year-old woman with rheumatoid arthritis was initiated on azathioprine 3 months ago and now presents with NAI-ALF secondary to DILI. On presentation, her ALT and AST concentrations are 500 IU/L and 350 IU/L, respectively. Her INR is 1.7, and she is mildly confused and drowsy. Which intervention has been shown most effective for the treatment of NAI-ALF?
   A. Intravenous acetylcysteine 21-hour regimen.
   B. Intravenous acetylcysteine 72-hour regimen.
   C. Oral acetylcysteine 72-hour regimen.
   D. Oral glutamine supplementation.

c. Viral hepatitis
   i. Hepatitis B
      (a) Similar to prophylaxis, acute hepatitis B infection treatment requires a nucleos(t)ide analog: entecavir 1 mg daily or lamivudine 100 mg daily.
   ii. Herpes simplex virus
      (a) Patients with HSV ALF can be treated with high dose acyclovir 5–10 intravenously every 8 hours for at least 7 days.

d. Mushroom poisoning
   i. Gastric lavage and activated charcoal may be beneficial for patients still having GI symptoms indicative of a recent ingestion.
   ii. Although data regarding efficacy are lacking, penicillin G or silibinin (silymarin or milk thistle) can be used as an antidote to α-amanitin, a toxin released after mushroom ingestion. Intravenous silibinin is guideline-preferred therapy; however, it is only available in United States through compassionate use, investigational drug access. Therefore, penicillin G is often used instead. Penicillin G directly competes with and inhibits the ability of the toxin to bind to plasma protein and penetrate the liver. Recommended dose of penicillin G for mushroom poisoning is 300,000 to 1 million units/kg/day given intravenously.

e. Autoimmune hepatitis
   i. High-dose intravenous corticosteroids (greater than or equal to 1 mg/kg prednisone equivalent or greater than or equal to 60 mg prednisone equivalent per day) should be initiated for patients who present with ALF secondary to autoimmune hepatitis. However, patients with progressive disease will require liver transplantation.

2. Management of neurologic complications
    a. Encephalopathy
        i. All medications that can cause sedation or confusion should be avoided (i.e., benzodiazepines, anticholinergics, etc.).
        ii. Grade I encephalopathy can typically be managed with close monitoring and without medication; grade II–IV encephalopathy should be treated in an ICU setting, if possible.
        iii. Given its ability to decrease serum ammonia concentrations and experience with treatment of hepatic encephalopathy in patients with cirrhosis, lactulose is recommended for patients with ALF with low-grade encephalopathy (e.g., grade I–II). The recommended dose is 20–30 g three or four times daily to produce 2 or 3 soft stools a day.
            (a) Despite its proposed benefits, retrospective data analyses of patients with ALF who receive lactulose therapy have not shown a benefit on encephalopathy or overall outcome.
            (b) In addition, lactulose has the potential to cause abdominal distension, which could be a concern for liver transplantation. Moreover, overuse of lactulose has the potential to cause intravascular depletion, which may further contribute to hemodynamic instability. Therefore, its effects may be harmful in the acute setting, particularly for patients with high-grade encephalopathy. In the 2023 ACG guidlines, the authors determine there is inconclusive evidence to recommend for or against the use of lactulose in the treatment of ALF.
        iv. Some centers use high-flow continuous renal replacement therapy (CRRT) to remove ammonia in patients with high-grade encephalopathy. CRRT has been associated with lower ammonia levels within the first 3 days of ALF (median reduction in ammonia is 38% with CRRT compared with 23% for intermittent RRT and 19% for no RRT). In the 2023 ACG guidelines, early CRRT was recommended for patients with grade II encephalopathy or higher.
        v. Patients with grade III and grade IV encephalopathy should be intubated for airway protection and treated with minimal sedation to allow for more frequent neurologic assessments. If sedation is necessary, propofol is typically used because it can reduce cerebral blood flow and lower ICP.
    b. Seizures
        i. Seizures have the potential to increase ICP. Therefore, seizures should be controlled quickly with short-acting benzodiazepines. If seizures persist, antiepileptic agents should be scheduled.
        ii. Use of prophylactic antiepileptics is not recommended. Studies have shown that use of prophylactic phenytoin in patients with ALF has no impact on prevention of seizures, cerebral edema, or overall survival.
    c. Elevated ICPs
        i. ICP should be kept less than 20–25 mm Hg while preserving CPP of at least 60 mm Hg.
        ii. Routine ICP monitoring has not been shown to reduce mortality in patients with ALF, and routine placement of ICP monitors is not recommended. Clinicians may choose to place an ICP monitor in patients with high-grade encephalopathy (grades III and IV) to provide close monitoring of cerebral edema. In addition, some centers may use noninvasive ICP monitoring strategies such as transcranial Doppler ultrasound to avoid the risks associated with more invasive ICP monitor placement in these coagulopathic patients. However, these monitoring strategies should be reserved for centers with large neurocritical care and neurosurgical experience.
        iii. Osmotic agents are used first line for control of ICP.
            (a) Mannitol has been used effectively in acutely reducing ICP in patients with ALF, though the effect is usually transient.
                (1) Mannitol is given as 0.5–1 g/kg intravenously once, which may be repeated to effect as long as the serum osmolality is less than 320 mOsm/L; however, mannitol is typically ineffective if the baseline ICP is greater than 60 mm Hg.

(A) Serum osmolality is estimated based on serum sodium, glucose, and blood urea nitrogen based on the following formula: (Na x 2) + (glucose/18) + (blood urea nitrogen/2.8).

(2) Adverse effects to consider for mannitol administration include fluid overload, particularly in patients with renal impairment, hyperosmolarity, hypotension, and hypernatremia.

(3) Hypertonic saline bolus (23.4% 30-mL bolus or 3% 200- to 300-mL bolus) is an alternative to mannitol for acute reductions in ICP. Adverse effects include hypernatremia and hyperchloremia.

(b) In patients with grade III or grade IV encephalopathy, multiorgan failure, or hemodynamic instability, prophylactic hypertonic saline (to goal 145–155 mEq/L) may be used to reduce the risk of cerebral edema.

(1) In a small, randomized controlled trial, 30 patients with ALF and grade III or grade IV encephalopathy were randomized to receive prophylactic hypertonic saline to maintain a serum sodium of 145–155 mEq/L compared with patients maintained at near-normal serum sodium levels (137–142 mEq/L). The primary outcome, incidence of ICP defined as elevations greater than 25 mm Hg, was significantly decreased in the hypertonic saline group (20% hypertonic saline vs. 46.7% control, p=0.04).

(2) Hypertonic saline in this study was administered as a 30% sodium chloride infusion by a syringe at 5–20 mL/hour; however, many preparation and dosing strategies have been used (e.g., 23.4% 30-mL bolus, 7.5% 2-mL/kg bolus, 3% 200- to 300-mL bolus, or continuous infusion), and the goal should be to target a serum sodium of 145–155 mEq/L.

iv. When severe ICP elevations do not respond to other measures, barbiturates such as pentobarbital may be used to control ICP.

(a) Profound hypotension may limit barbiturate use in ALF when patients have hemodynamic instability at baseline. Patients may require vasopressors to maintain adequate MAP (and CPP) while receiving barbiturates.

(b) Barbiturate clearance is significantly decreased in patients with ALF, which may limit clinicians' ability to perform neurological assessments for extended periods.

v. Hyperventilation to a $Paco_2$ of 25–30 mm Hg can restore cerebral autoregulation, which results in vasoconstriction and decreased ICP.

(a) The effects of hyperventilation on ICP appear to be short-lived. A randomized controlled trial of prophylactic hyperventilation showed no benefit on cerebral edema and survival. In addition, there is concern that cerebral vasoconstriction with hyperventilation may worsen cerebral hypoxia.

(b) Thus, hyperventilation currently plays no role in prevention of ICP and should only be considered for refractory treatment of acute ICP.

vi. Hypothermia (33°C–34°C) may control ICP in patients with ALF by lowering the production of ammonia and by decreasing the cerebral uptake of ammonia and cerebral blood flow. However, hypothermia for patients with ALF has not been compared with normothermia in controlled trials, and a recent retrospective cohort study showed no difference in overall and transplant-free survival when compared with normothermia. In addition, there are concerns about coagulation disturbances and increased risk of infection with hypothermia.

Patient Case

2. A 33-year-old man presents with ALF secondary to acetaminophen overdose. He is now 72 hours post-ingestion and is profoundly encephalopathic and unresponsive to pain on examination. An ICP monitor is placed, which shows acute elevations of 30 mm Hg. Which is most appropriate for the acute management of ICP elevations?
   A. Hypertonic saline continuous infusion to maintain serum sodium 145–155 mEq/L.
   B. Mannitol 0.5 mg/kg intravenously × 1.
   C. Hyperventilation to $Paco_2$ of 25–30 mm Hg.
   D. Thiopental continuous infusion.

3. Management of hemodynamic instability
   a. Patients should initially be resuscitated with 0.9% sodium chloride. Hypo-osmolar fluids including lactated Ringer solution (273 mOsm/L) should be avoided, if possible, in patients with grade III and grade IV encephalopathy because of the risk of cerebral edema. In addition, Plasma-Lyte (294 mOsm/L) could be used as an alternative to 0.9% sodium chloride (308 mOsm/L), given that it is not hypo-osmotic and would provide a balanced solution to avoid the risk of hyperchloremic metabolic acidosis.
   b. Vasopressors should be used if fluid resuscitation fails to maintain a MAP greater than 75 mm Hg or a CPP of at least 60 mm Hg.
      i. Norepinephrine is the vasopressor of choice in patients requiring vasopressor support.
      ii. Use of vasopressin is controversial for patients with ALF who have high-grade encephalopathy. One study of six patients with grade IV encephalopathy showed an increased ICP after 1 hour of terlipressin (a synthetic vasopressin analogue), though systemic hemodynamics were not significantly altered; however, a subsequent study did not show similar effects on ICP with terlipressin. Because there have been no similar studies of vasopressin in patients with ALF, its use in patients with ALF and high-grade encephalopathy should be cautioned. However, in the most recent Society of Critical Care Medicine (SCCM) supportive care management guidelines for patients with acute liver failure as well as the 2023 ACG ALF guidelines, vasopressin is recommended to be added to norepinephrine with persistent shock, despite these concerns, although both guidelines note a low level of evidence.
4. Management of coagulopathies
   a. Although the INR may be elevated in patients with ALF, overall hemostasis is maintained through compensatory mechanisms.
   b. Because traditional measures of coagulation may be unreliable, guidelines suggest use of viscoelastic testing such as thromboelastography over measuring INR, platelets, and fibrinogen when available. In a small randomized trial of 60 cirrhotic patients, blood product transfusion was significantly reduced when guided by thromboelastography compared with traditional measures with no differences in bleeding complications. Unfortunately, these tests are not available at many institutions and traditional measures are often still required.
   c. In patients without acute bleeding or high-risk procedure such as ICP monitor placement, the routine correction of coagulopathy is not supported. For instance, in patients with an elevated INR without signs and symptoms of an acute bleed, INR should not be corrected using fresh frozen plasma. Vitamin K (5–10 mg) may be administered because many patients with ALF are deficient in vitamin K, further contributing to the coagulopathy of ALF. Intravenous administration is usually recommended because subcutaneous administration of vitamin K can lead to erratic absorption, and enteral absorption may also be unreliable.

d. If clinically significant bleeding occurs, INR correction with fresh frozen plasma is warranted.
   i. Guidelines recommend an INR correction to about 1.5 for clinically significant bleeding.
   ii. If a fresh frozen plasma infusion alone does not adequately lower INR, recombinant activated factor VII (rFVIIa) may be administered. In a small nonrandomized study of 15 patients with fulminant hepatic failure, administration of rFVIIa at a dose of 40 mcg/kg temporarily improved coagulation parameters (i.e., INR less than 1.6) compared with patients receiving only fresh frozen plasma (100% vs. 0%, $p<0.002$). Although the improvements in coagulation were only temporary, patients in the rFVIIa group were able to have invasive procedures performed (e.g., ICP monitors placed) more often than were patients in the control group (100% vs. 38%, $p=0.03$).
   iii. Data are limited on use of prothrombin complex concentrate (PCC) for reversing coagulopathy in ALF; however, it appears to be similar in efficacy to rFVIIa. A retrospective study in patients with cirrhosis found similar rates of INR reversal (1.5 or less) in patients who received either PCC (80%; $p=0.03$) and rFVIIa (87%; $p=0.04$), compared with FFP (27%). Therefore, PCC may be an option in cases for which the goal is to limit volume associated with FFP; however, INR-based dosing limits this use, and further study is warranted.

e. To reduce the risk of spontaneous intracranial hemorrhage, platelet transfusions should be provided if the count drops to less than 15,000–20,000/mm^3 in the absence of bleeding.
   i. If clinically significant bleeding occurs, patients should be transfused to a target platelet count greater than 50,000/mm^3.
   ii. For invasive procedures, platelet counts should be 50,000–70,000/mm^3 to prevent bleeding, though thromboelastography data analyses suggest that a target of 100,000/mm^3 is ideal.

f. Patients with ALF should be evaluated for venous thromboembolism prophylaxis based on traditional risk assessments (e.g., thrombocytopenia, recent and/or active bleeding, etc). Venous thromboembolism prophylaxis should not be withheld due to elevated INR because INR often overestimates hypercoaguability in patients with ALF. Selection of agent for prophylaxis is unclear. No studies have compared low-molecular-weight heparin with unfractionated heparin in patients with ALF; however, most studies of patients with cirrhosis support the use of LMWH, though the control group is no-prophylaxis.

g. Histamine-2 receptor antagonists or PPIs should be initiated in patients with ALF to reduce the incidence of spontaneous GI bleeding.

5. Management of infectious complications
   a. Antimicrobial prophylaxis is often given because infection remains the primary cause of death in patients with ALF. However, data analyses are limited on the benefit of antimicrobial prophylaxis in ALF. A retrospective cohort study of 1551 patients with ALF showed that prophylactic antibiotic therapy did not reduce the incidence of bloodstream infections (12.8% in the prophylaxis group vs. 15.7% in the non-prophylaxis group, $p=0.12$) and did not reduce 21-day mortality.
   b. Patients should be monitored closely for infection through surveillance cultures, and antimicrobials should be initiated promptly if the patient has any signs or symptoms of a systemic infection. Empiric administration is recommended for any of the following scenarios:
      i. Positive surveillance cultures
      ii. Progression to higher-grade encephalopathy
      iii. Refractory hypotension
      iv. Development of systemic inflammatory response syndrome (SIRS) criteria

G. Advanced Therapies for ALF
   1. Liver assist devices: The molecular adsorbent recirculating system (MARS) has been studied for the treatment of ALF. In a large propensity-matched study of 104 patients who received MARS for ALF (matched 4:1 to 416 controls), MARS significantly increased the odds of 21-day transplant-free survival (OR 1.9; 95% CI, 1.07–3.39; p=0.03).
   2. Transplantation
      a. Orthotopic liver transplantation is the only definitive treatment for patients with ALF.
      b. The advances in liver transplantation have improved ALF survival to greater than 60%.

H. Prognosis
   1. Predictive models of survival and need for liver transplantation are useful to identify the patients most likely to benefit from liver transplantation.
   2. The most important predictor of outcome seems to be the cause of ALF. Transplant-free survival is 50% or greater when acetaminophen, hepatitis A, acute ischemic injury, or pregnancy is the cause, whereas other causes confer less than 25% transplant-free survival. In addition, transplant-free survival is significantly decreased when patients present with grade III or grade IV encephalopathy compared with grade I or grade II encephalopathy.
   3. King's College criteria, developed from a cohort of about 600 patients with ALF, incorporate parameters such as cause of ALF and clinical parameters, including degree of encephalopathy and liver function tests, in order to evaluate the decision to perform transplantation versus provide medical therapy.

II. ACUTE PANCREATITIS

A. Epidemiology
   1. AP is responsible for 275,000 hospitalizations per year in the United States.
   2. In 2012, AP was the third most common gastroenterology diagnosis in the United States, with an annual cost of $2.6 billion.
   3. Incidence is estimated at 5–30 cases per 100,000.
   4. AP is classified according to disease severity. Overall, about 20% of patients with AP have a severe course with an overall mortality rate of approximately 5%.

B. Definitions
   1. The clinical diagnosis of AP is based on characteristic symptoms (i.e., abdominal pain and nausea), together with elevated serum concentrations of pancreatic enzymes. According to the Acute Pancreatitis Classification Working Group definition, two of the following three criteria must be met for a patient to be given a diagnosis of AP:
      a. Abdominal pain consistent with AP. This is typically persistent epigastric pain, sometimes radiating to the back.
      b. Serum lipase (or amylase) concentrations greater than 3 times the upper limit of normal
      c. Imaging CT, magnetic resonance imaging [MRI], or transabdominal ultrasonography) consistent with pancreatitis
   2. AP is classified according to disease severity as mild, moderately severe, or severe.
      a. Mild AP: Tends to be self-limiting with no organ failure or necrosis. There is also the absence of local and systemic complications.
      b. Moderately severe AP: Characterized by local complications and organ failure that lasts less than 48 hours. Local complications for moderately severe AP resolve without intervention.

   c. Severe AP: Characterized by persistent organ failure for more than 48 hours. Patients with severe AP usually have one or more local complications. Mortality rates are highest for severe AP (up to 20% of cases), particularly if persistent organ failure develops within the first few days of disease.
3. AP is further classified according to the phase. There are two distinct phases in this disease: early and late. The early phase usually lasts for the first week, and the late phase can last for weeks to months:
   a. Early: Symptoms of SIRS (e.g., temperature greater than 38°C or less than 36°C, heart rate greater than 90 beats/minute, respiratory rate greater than 20 breaths/minute or $PaCO_2$ greater than or equal to 32 mm Hg or ventilator use, WBC greater than 12,000 or less than 4,000) with or without transient organ failure during the first week. Inflammatory responses and organ failure are mainly because of a response to local pancreatic injury.
   b. Late: Complications that occur after 1 week that can last weeks to months. This phase is characterized by persistent organ failure and the presence of local complications. The late phase occurs only in those with moderately severe or severe AP.
4. Local complications with moderately severe and severe AP may include:
   a. Peripancreatic fluid collections
   b. Pancreatic and peripancreatic necrosis (sterile or infected). Pancreatic necrosis is an area of nonviable pancreatic parenchyma identified by lack of enhancement on imaging, generally involving greater than 30% of the pancreas.
   c. Pseudocysts
   d. Walled-off necrosis (sterile or infected)
5. Organ failure associated with AP is typically cardiovascular, respiratory, or renal dysfunction. The modified Marshall scoring system is recommended to identify patients with AP with organ failure. A score of 2 or more for one of these three systems indicates organ failure.

C. Pathophysiology – AP is primarily caused by inappropriate activation of trypsinogen to trypsin. Trypsin is the key enzyme responsible for the activation of pancreatic zymogens. When trypsin is inappropriately formed and retained in the pancreas, activation of digestive enzymes inside the pancreas causes pancreatic autodigestion and pancreatic injury.

D. Causes
1. Gallstones
   a. Gallstone pancreatitis is the main cause of AP, responsible for 40%–70% of cases; 3%–7% of patients with gallstones develop pancreatitis.
   b. Gallstone pancreatitis occurs primarily in white women older than 60 years and in patients with small stones (less than 5 mm in diameter).
   c. Gallstone pancreatitis is usually an acute event that resolves once the gallstone has been removed or has passed.
   d. Patients may undergo a cholecystectomy to prevent further episodes in the future.
2. Alcohol use
   a. Excessive alcohol use is the etiology of pancreatitis in 25%–35% of cases.
   b. Symptoms can occur as an acute episode or present as a chronic pancreatitis.
   c. Alcohol-induced pancreatitis is more common in men than in women.
   d. Alcohol use and its association with pancreatitis is thought to have a dose-dependent relationship. Alcohol intake for more than 5 years or greater than 50 g daily increases the risk of pancreatitis. However, only about 5% of patients with a history of heavy alcohol consumption develop AP.

3. Hypertriglyceridemia
    a. Primary and secondary hypertriglyceridemia are increasingly recognized as causes of AP (5%–22% reported).
    b. If serum triglycerides are greater than 1000 mg/dL, hypertriglyceridemia should be suspected as the cause.
4. Certain medications can cause pancreatitis through varied mechanisms. Drug-induced pancreatitis is classified based on level of evidence and number of cases reported (Class 1, high level of evidence, through Class 4, very low quality of evidence). Examples include:
    a. Angiotensin-converting enzyme (ACE) inhibitors (Class 2)
    b. Azathioprine, 6-mercaptopurine (Class 1)
    c. HAART (highly active antiretroviral therapy) medications (e.g., didanosine) (Class 1, didanosine)
    d. DPP4 inhibitors, GLP1 agonists (Class 2)
    e. Valproic acid (Class 2)
    f. Propofol (Class 4)
5. Malignancy – The presence of a pancreatic tumor blocking the main pancreatic duct should be suspected in any patient older than 40 years with signs and symptoms of pancreatitis and no other apparent cause.
6. About 20% of AP cases are idiopathic.

E. Diagnosis
1. Signs and symptoms
    a. Abdominal pain
        i. AP pain is usually located in the epigastric region or left upper quadrant; however, it may radiate to the back, chest, or flank. Pain is typically constant and severe.
        ii. Gallstone pancreatitis–induced pain has been described as knifelike.
    b. Nausea
2. Laboratory abnormalities
    a. According to the Acute Pancreatitis Classification Working Group definition of AP, serum lipase (or amylase) concentrations at least 3 times the upper limit of normal are required for diagnosis.
    b. Serum lipase is preferred for diagnosis because elevations in serum lipase concentrations are more specific to the diagnosis of AP than elevations in amylase and serum lipase concentrations remain elevated longer than amylase concentrations.
        i. Amylase concentrations rise quickly (within a few hours) in AP; however, they return to normal within a few days.
        ii. Amylase concentrations may remain normal in alcohol-induced AP and hypertriglyceridemia.
3. Imaging
    a. Transabdominal ultrasonography should be done to confirm the diagnosis of AP for all patients.
    b. Contrast-enhanced CT scans of the abdomen are more than 90% sensitive and specific in diagnosing AP; however, routine use is unnecessary.
    c. Contrast-enhanced CT and/or MRI should be used in patients with an unclear diagnosis or in patients who do not improve after 48–72 hours to evaluate for complications.

F. Management
1. Hydration
    a. Patients with AP need early aggressive volume resuscitation because they are volume depleted for many reasons, including vomiting, reduced oral intake, third spacing of fluids because of inflammatory response, and diaphoresis. In fact, it has been hypothesized that worsening pancreatic hypoperfusion that develops from pancreatic inflammation in the setting of volume depletion leads to pancreatic necrosis. Thus, early aggressive volume resuscitation may prevent the development of pancreatic necrosis.

b. Early aggressive volume resuscitation (i.e., within the first 24 hours) has been shown to be more effective than later aggressive volume resuscitation. Studies that have continuously used aggressive hydration strategies beyond the first 24 hours of presentation of AP have not shown benefit.
   i. Although early aggressive volume resuscitation is recommended by the guidelines, this approach is defined in several different ways, including either 5–10 mL/kg/hr or 250–500 mL/hr for the first 12– 24 hours. In addition, the 2018 American Gastroenterological Association (AGA) guidelines suggest using either noninvasive (e.g., heart rate, mean arterial pressure, urine output) or invasive measures (e.g., stroke volume variation) to determine fluid responsiveness.
   ii. A 2022 randomized controlled trial compared aggressive fluid resuscitation (defined as 20 mL/kg bolus followed by 3 mL/kg/hr) with moderate fluid resuscitation (defined as 1.5 mL/kg/hr for all patients, with hypovolemic patients receiving an initial 10 mL/kg bolus). Of note, all study patients were diagnosed with acute pancreatitis, and the outcome was progression to moderately severe or severe pancreatitis. The study was stopped early, however, because of increased harm in patients who received aggressive resuscitation, primarily fluid overload (20.5% vs. 6.3%; p=0.004) without any differences in efficacy outcomes. This finding suggests that less aggressive fluid strategies may be warranted, particularly for patients who present with less severe pancreatitis. However, in patients who present with severe AP, more rapid initial aggressive volume resuscitation may still be warrented.

c. Selection of intravenous fluid for initial volume resuscitation in AP may be important, favoring pH-balanced fluids. Low pH activates trypsinogen, implying that low-pH fluids exacerbate inflammatory response in AP. In a randomized controlled trial comparing initial resuscitation (within the first 24 hours) with lactated Ringer solution to 0.9% sodium chloride for patients with AP, there was a significant reduction in SIRS after 24 hours in patients resuscitated with lactated Ringer compared with 0.9% sodium chloride (84% reduction vs. 0%, respectively, p=0.035). The study also showed a significant reduction in C-reactive protein concentrations with administration of lactated Ringer solution compared with 0.9% sodium chloride. However, two recent randomized controlled trials comparing resuscitation with lactated Ringer solution and 0.9% sodium chloride showed similar early (within first 24 hours) reductions in SIRS with lactated Ringer solution, but no differences in SIRS prevalence after 24 hours or clinical outcomes, including mortality. Thus, because no study has shown meaningful clinical differences in fluid selection, the 2018 AGA guidelines make no recommendation on intravenous fluid of choice.

d. Because there can be harm in over-resuscitating patients, especially those with concomitant cardiovascular or renal failure, aggressive hydration may not be beneficial if there is no response within the first 6–12 hours.

2. Etiology-specific treatments
   a. Insulin for hypertriglyceridemia-induced AP
      i. High-dose insulin infusion is the treatment mainstay for severe AP secondary to hypertriglyceridemia.
      ii. Insulin is initiated at 0.1 units/kg/hour, with dose adjusted on the basis of serum triglyceride concentrations, which are typically checked every 12 hours. If serum triglycerides do not decrease by at least 25%–50% in 24 hours, the insulin infusion is increased by 0.05 units/kg/hour until the goal serum triglyceride is achieved (ideally less than 500 mg/dL).
      iii. To prevent hypoglycemia with high-dose insulin, blood glucose (BG) should be monitored hourly, and a dextrose infusion should be initiated and adjusted to maintain the BG at around 150 mg/dL.

b. Endoscopic retrograde cholangiopancreatography (ERCP)
   i. Although many gallstones are passed through the duodenum and lost in the stool without causing harm to the patient, gallstones that are not cleared can cause an obstruction in some patients in either the biliary tree or the pancreatic duct, which can lead to severe AP or cholangitis.
   ii. An ERCP, together with a sphincterotomy, may be used to extract gallstones from the pancreatic ducts or biliary tree.
   iii. There is considerable risk with ERCP, including bleeding and the potential to worsen AP because of manipulation of the pancreas.
   iv. Eight randomized controlled trials have evaluated the effectiveness of early ERCP in reducing the risk of complications in patients with gallstone AP. A limitation of these studies is that many did not exclude acute cholangitis, which is an indication for ERCP without AP. A systematic review controlling for patients with cholangitis found no benefit of early ERCP (mortality OR 0.67; 95% CI, 0.26–1.75). Given these limitations and results, the current guidelines do not recommend routine use of ERCP in patients without cholangitis.
   v. AP is the most common complication of ERCP. Although the incidence of post-ERCP pancreatitis has decreased significantly during the past few decades, post-ERCP pancreatitis continues to occur in 2%–4% of patients.
      (a) Rectal NSAIDs (i.e., 100 mg of indomethacin) after an ERCP can be given to patients at high risk of post-ERCP pancreatitis.
      (b) In a multicenter randomized controlled trial of 600 high-risk patients undergoing ERCP, 100 mg of rectal indomethacin (administered as two 50-mg suppositories given immediately post-ERCP) reduced the incidence of AP compared with placebo (9.2% vs. 16.9%, p=0.005) and, specifically, the development of moderate to severe AP (4.4% vs. 8.8%, p=0.03).

3. Infection
   a. Pancreatic and non-pancreatic infections contribute to mortality in patients with AP. The primary infectious concern with AP is infected necrotizing pancreatitis, which confers a high mortality rate (about 30%); thus, patients with severe acute necrotizing pancreatitis are at highest risk of pancreatic infections.
   b. The role of prophylactic antibiotics is controversial in AP.
      i. Early randomized trials showed a potential benefit in the reduction of infectious complications, particularly central line–related bloodstream infections, pulmonary infections, urinary infections, and pancreatic infections; however, the studies were unblinded, and the benefit was mostly confined to patients with severe AP.
      ii. In more recent literature with higher-quality trials, the benefit of prophylactic antibiotic is unsupported. A systematic review of 10 randomized controlled trials showed a trend toward reduced mortality with prophylactic antibiotics (OR 0.66; 95% CI, 0.42–1.04); however, the mortality benefit was lost among the subgroup of recent studies (after 2002: OR 0.85; 95% CI, 0.52–1.80). Similarly, there was no difference in risk of infected necrosis in more recent trials (OR 0.81; 95% CI, 0.44–1.49).
   c. According to the available information, use of prophylactic antibiotics to prevent the development of infected pancreatic necrosis is not recommended at this time.
   d. For patients who have not improved after 7–10 days of hospitalization with pancreatic necrosis, an infection should be suspected, and empiric antibiotics should be initiated at that time. Because of penetration issues, carbapenems, fluoroquinolones, metronidazole, or high-dose cephalosporins may be preferred; however, guidelines now also recommend piperacillin-tazobactam for severe AP, despite intermediate penetration..

4. Nutrition
    a. Historically, patients were kept NPO in order to rest the pancreas and prevent the further release of pancreatic enzymes. Patients were given TPN until complete resolution of AP.
        i. This practice has largely fallen out of favor because several studies have shown that early enteral nutrition is safe and effective for patients with AP. In addition, a recent meta-analysis of 11 randomized trials concluded that early (within 24–48 hours) feedings were not associated with increased adverse effects and may reduce hospital length of stay, particularly for mild to moderate AP. Enteral feeding maintains the gut mucosal barrier and helps prevent infectious complications such as infected necrosis, which may result from bacterial translocation from the gut.
        ii. A meta-analysis of eight trials comparing enteral nutrition with TPN in patients with AP showed decreased mortality (relative risk [RR] 0.5; 95% CI, 0.28–0.91), infectious complications (RR 0.39; 95% CI, 0.23–0.65), multiorgan failure (RR 0.55; 95% CI, 0.37–0.81), and surgical interventions (RR 0.44; 95% CI, 0.29–0.67) with the use of enteral nutrition. Enteral nutrition in these trials was given by the NJ route.
        iii. TPN is mainly reserved for patients unable to meet caloric demands with enteral feeding.
    b. Although the NJ route for enteral feeding has been preferred because it avoids the gastric area where pancreatic enzyme stimulation may occur, the NG route appears to be safe.
        i. A recent meta-analysis of three randomized controlled trials showed no significant differences in mortality, tracheal aspiration, or proportion meeting energy balance between the two routes; however, the results were limited by small sample sizes.
        ii. NG feeding is easier than NJ feeding because NJ tubes can be difficult to place, expensive, and inconvenient. However, there is concern that NG feeding may increase the risk of aspiration pneumonitis, particularly in patients with a history of aspiration.
5. Surgery
    a. Cholecystectomy should be performed in patients with gallstones retained in the gallbladder to prevent the recurrence of AP. For necrotizing biliary AP, cholecystectomy should be delayed until inflammation is resolved and fluid collections are cleared in order to avoid infection.
    b. Surgery is generally unnecessary for asymptomatic patients with pseudocysts and pancreatic or extrapancreatic necrosis.
    c. Surgical debridement of sterile necrosis is only necessary if gastric outlet obstruction or bile duct obstruction is present.
    d. Surgical intervention is the treatment of choice for infected necrotizing pancreatitis.
        i. For stable patients with infected necrotizing pancreatitis, surgical debridement should be delayed for at least 4 weeks to allow appropriate delineation of necrotic versus non-necrotic tissue, and antibiotics should be tried before surgical intervention.
        ii. Unstable patients with infected necrosis should undergo immediate debridement, and necrosectomy may be required in patients who do not respond to a combination of antibiotics and debridement.

Patient Case

3. A 44-year-old woman presents to the ED with a 2-day history of diffuse abdominal pain. Amylase is 700 IU/L, and lipase is 800 IU/L. She is given intravenous fluids with lactated Ringer solution and enteral nutrition through an NJ tube. However, after 48 hours, she has still not improved, and CT imaging reveals pancreatic necrosis involving 40% of the pancreas. Which interventions would best be performed at this time?
   A. Piperacillin/tazobactam should be prophylactically administered to prevent infected necrotizing pancreatitis.
   B. Surgical management of necrotizing pancreatitis is necessary.
   C. Antibiotics should be deferred unless systemic signs of infection are present.
   D. Meropenem should be prophylactically administered to prevent infected necrotizing pancreatitis.

III. GASTROINTESTINAL FISTULAS

A. Epidemiology
   1. About 20% of patients with Crohn disease will develop a spontaneous fistula in their lifetime, and up to 12% of patients with diverticulitis will develop a spontaneous fistula.
   2. Incidence of postoperative fistula formation varies from less than 1% to about 35%, depending on the type of abdominal surgery, with the rate of fistula increasing with more complex surgical procedures and more complicated resections/anastomoses. Additional factors such as Crohn's disease (incidence 15%–35%) increases the risk of fistula formation postoperatively.
   3. In the past 50 years, the mortality rate associated with fistulas has decreased significantly from greater than 40% to currently around 20%, mainly because of improvements in supportive care and the advent of nutritional support.
   4. GI fistulas remain a source of considerable morbidity with prolonged hospital courses, infections, and malnutrition.

B. Definition
   1. A fistula is any abnormal connection between two epithelialized surfaces. GI fistulas involve an abnormal connection between the GI tract and the skin, another internal organ, or an internal cavity such as the peritoneal or pleural space.
   2. Classification systems for fistulas:
      a. Anatomic: Describe the fistula origin and drainage point
         i. Internal: Connection to another internal organ or internal cavity (i.e., ileocolic)
         ii. External: Connection to the skin (i.e., enterocutaneous)
      b. Nomenclature: Typical method is from point of origin (e.g., gastro-, entero-, ileo-, etc.) followed by point of termination (e.g., -cutaneous, -colic, etc.).
      c. Physiologic: Classified according to daily fistula output. Daily fistula output is one of the most important determinants of morbidity and mortality, and it generally decreases before spontaneous closure.
         i. High output: Greater than 500 mL/day. High-output fistulas are associated with a mortality rate of about 35%.
         ii. Moderate output: 200–500 mL/day
         iii. Low output: Less than 200 mL/day

d. Fistulas can also be described according to whether they maintain continuity with the GI tract.
    i. Lateral fistulas divert off the intestines while maintaining the continuity of the intestinal tract. With lateral fistulas, intestinal contents follow normal progression beyond the fistula.
    ii. End fistulas disrupt the continuity of the intestinal tract beyond the fistula.

C. Causes
1. Postoperative fistulas
    a. Most fistulas are formed after surgery (about 80%), most commonly after operations for cancer.
    b. Postoperative fistulas form because of either infection or breakdown at intestinal anastomoses.
    c. Postoperative fistulas are more commonly external, often because of the presence of a drain.
2. Spontaneous fistulas
    a. 15%–25% of GI fistulas occur spontaneously.
    b. Crohn disease and inflammatory bowel disease are the leading cause of spontaneous fistula formation, though pancreatitis and cancer (particularly if radiation therapy is involved) can also lead to spontaneous fistulas.
    c. Spontaneous fistulas often form because of local inflammatory processes that can cause local abscess formation or perforation.
    d. Spontaneous fistulas can be either external or internal.
3. Trauma-induced fistulas
    a. Some fistulas are caused by trauma (abdominal wounds or blunt trauma).
    b. Trauma may lead to fistula formation by causing vascular injury.

D. Diagnosis
1. Common presenting symptoms include pain, abdominal tenderness, leukocytosis, and fever. External fistulas are generally easier to recognize because of the presence of effluent at the drainage sites. Patients with internal fistulas may present with nonspecific symptoms such as diarrhea, dyspnea, or SIRS.
2. In postoperative patients, the first indication of a fistula is delayed recovery, which usually occurs within the first week after surgery.
3. Diagnostic workup should include fistula output volume, fistula output description (e.g., color, consistency), biochemical evaluation of fistula content (e.g., water-electrolyte balance, amylase, lipase, pH), microbiological evaluation of fistula content, and markers of nutritional status.
4. Radiographic contrast studies using CT or MRI are necessary to determine the anatomic aspects of the fistula (e.g., origin, length of fistula, presence of obstruction or abscesses). Barium is generally used for contrast because of its ability to show mucosal surfaces.

E. Treatment
1. Fluid resuscitation and electrolyte management
    a. GI fistula fluid is typically iso-osmotic and rich in sodium, potassium, chloride, and bicarbonate. High-output fistulas can result in large fluid and electrolyte imbalances leading to dehydration, hypokalemia, hyponatremia, and metabolic acidosis.
    b. Fistula output from the upper GI tract is generally more acidic and rich in potassium. Replacement fluid should include a maintenance infusion of 0.9% sodium chloride with potassium and frequent reassessments for potassium corrections.
    c. Pancreatic and duodenal fistulas result in bicarbonate losses and may require bicarbonate replacement.
    d. Fistula fluid composition may be analyzed in order to correctly replete fluid and electrolyte deficits.

2. Drainage
    a. Drainage is used to prevent the accumulation of fluid and the development of infection.
    b. Most enterocutaneous fistulas will drain to skin spontaneously; however, some fluid may be retained in the fistula tract, which may lead to infection.
    c. Vacuum-assisted closure (VAC) devices administer negative pressure wound therapy and can increase blood flow and decrease fluid collections. For enterocutaneous fistulas, a VAC system can help protect skin and decrease fistula output.
3. Nutrition
    a. Nutritional deficiencies are present in 55%–90% of patients with GI fistulas, particularly with upper GI fistulas, because there is substantial fluid, electrolyte, and protein loss from the upper GI tract.
    b. Because of nutritional deficits, patients may need nutritional supplementation in excess of daily demands. Patients with low-output fistulas may need additional protein, and patients with high-output fistulas may need additional daily caloric and protein requirements (e.g., 1.5–2 times basal daily expenditure).
    c. Enteral feedings are the preferred method for nutritional supplementation because enteral feedings provide direct stimulation to the enterocyte, which may enhance mucosal proliferation.
        i. In a retrospective cohort study of 335 patients with high-output small intestine fistulas (median output 1350 mL/day), 85% (n=285) were treated with enteral nutrition. Spontaneous closure rates were acceptable.
        ii. Tolerability may limit the use of enteral feeding (e.g., high gastric residuals, diarrhea).
    d. TPN has the potential advantage of improving spontaneous closure rates because of reductions in GI secretions and reduction in fistula output volumes, particularly for patients with high-output fistulas and when combined with antisecretory agents (e.g., octreotide).
    e. Fistula site may also influence the choice of enteral nutrition compared with TPN. In general, enteral feeding is provided for fistulas from the esophagus, lower small intestine, and colon, whereas TPN is used for fistulas from the stomach, pancreas, or upper small intestine.
    f. Some evidence supports adding supplements to enteral feeding. These include fish oil, omega-3 fatty acids, and glutamine, which may improve immune function or increase blood flow to the intestinal tissue. In a study of 28 patients with high-output fistulas, patients who received glutamine supplementation (0.3 g/kg/day orally) in addition to TPN were significantly more likely to have spontaneous fistula closure.
4. Somatostatin and octreotide
    a. Somatostatin is a tetradecapeptide that is naturally found in the GI tract and the nervous system.
        i. It has several biological effects, including inhibition of hormone secretion (i.e., gastrin, cholecystokinin, secretin, insulin, glucagon), inhibition of exocrine secretory response (i.e., gastric acid and pancreatic secretion), inhibition of motor activity in the GI tract, inhibition of nutrient absorption, and stimulation of water and electrolyte absorption.
        ii. Octreotide is an octapeptide synthetic analog of somatostatin with similar activity.
        iii. Because output volume is correlated with spontaneous closure, drugs such as somatostatin and octreotide are used to reduce output volumes.
    b. Somatostatin vs. octreotide
        i. Somatostatin has a very short half-life (about 1–2 minutes), which requires a continuous infusion, whereas octreotide has a half-life of around 2 hours, allowing for intermittent dosing (e.g., three times daily).
        ii. Effect of octreotide appears to diminish with repeated dosing, possibly because of down-regulation of somatostatin receptors.
        iii. Somatostatin is active at all somatostatin receptors, whereas octreotide has variable affinity at the somatostatin receptors.

c. Somatostatin significantly reduces output volumes. One prospective, randomized controlled single-center trial compared somatostatin 250 mcg/hour intravenously continuously with placebo for patients receiving TPN. Somatostatin significantly reduced the time to achieve a 50%, 75%, and 100% reduction in fistula output compared with placebo; also, although there was no difference in the rates of fistula closure (85% vs. 81.25%), the time to fistula closure was significantly reduced with the use of somatostatin (13.9 days vs. 20.4 days, $p<0.05$).
d. Octreotide has been shown to decrease fistula output in some studies, though in other trials, it had no effect on fistula output.
    i. One small study of 14 patients showed a beneficial effect of octreotide on output volumes. In this crossover study, octreotide at 100 mcg subcutaneously three times daily significantly reduced fistula output compared with placebo for the first 2 days of therapy by about 400 mL/day. When the group that was originally randomized to receive octreotide crossed over to the placebo arm, output increased by about 250 mL/day.
    ii. Two subsequent studies did not show similar results on fistula output. Possible reasons for decreased efficacy with octreotide include diminished response with repeat dosing and decreased activity at some somatostatin receptors.
e. Reduction in fistula output with the use of somatostatin or octreotide should occur within 48 hours. If no noticeable response in fistula response occurs at 48 hours, treatment should be discontinued.

5. Refractory high-output fistula management
    a. Acid-suppressing medications (e.g., PPIs and histamine-2 receptor antagonists) have been studied for the treatment of refractory fistula because of their ability to decrease gastric acidity and decrease the amount of gastric secretions. Proton pump inhibitors are more effective than histamine-2 receptor antagonists for refractory fistula output.
    b. Antimotility agents are usually recommended (e.g., loperamide, diphenoxylate/atropine, codeine) because of their ability to inhibit the activity of gastrointestinal tract muscles.
        i. Loperamide has the greatest effect on fistula output. In studies, loperamide doses up to 12–24 mg per dose have been given safely for the management of high-output fistulas. However, due to concerns for adverse reactions, caution should be advised with doses higher than 16 mg per day. Additionally, administration of large quantities of the liquid form of loperamide may increase fistula output.
6. Conservative versus surgical management
    a. Conservative management is first line for most patients, with the primary goal being spontaneous closure of the fistula. Several prognostic indicators improve the likelihood of spontaneous closure.
        i. Low-output fistulas
        ii. Patients younger than 40 years
        iii. Fistula sites: Oropharyngeal, esophageal, duodenal, pancreatic, jejunal
        iv. A long fistula tract (greater than 2 cm)
        v. Intestinal continuity maintained
    b. Surgery is usually indicated for fistulas that fail to close spontaneously after 30–60 days. Some causes of fistula formation (e.g., bowel injury caused by trauma or certain surgical procedures) may require emergency surgery to repair damage.

Patient Case

4. A 34-year-old woman with morbid obesity presents to the surgical ICU after gastric bypass surgery 1 week prior with an enterocutaneous fistula requiring medical management. Her current output is about 600 mL/day. For the patient's high-output fistula, which best represents the intervention that has not been associated with a reduction in fistula output volume?
   A. Somatostatin 250 mcg/hour intravenously continuously.
   B. Octreotide 100 mcg subcutaneously three times daily.
   C. TPN.
   D. Glutamine 0.3 g/kg/day orally.

IV. POSTOPERATIVE ILEUS

A. Epidemiology
   1. Incidence of POI can vary, depending on the type of procedure:
      a. Abdominal hysterectomy: About 3%
      b. Bowel resection: About 15%
   2. POI can lead to a prolonged hospital stay, prolonged recovery, and increased morbidity including increased postoperative pain, PONV, and risk of postoperative complications (e.g., aspiration pneumonia, thromboembolism, nosocomial infection).

B. Definition
   1. POI is a transient impairment of appropriate GI motility after a surgical procedure.
   2. The paralytic state in POI is not caused by a mechanical obstruction, and ileus can affect the stomach, small intestine, or large intestine.
   3. The duration of POI is typically 2–3 days after a procedure, but POI may last up to 6 days postoperatively. Return to normal bowel function is monitored using objective signs such as passing of flatus, active bowel sounds, or a bowel movement.
      a. The duration of POI often depends on the surgical site. Return to normal function is fastest for the small bowel, normally within 24 hours. Paralytic state may last on average 24–48 hours in the stomach, whereas it may take up to 3–5 days for the colon to return to normal function.
      b. If POI persists beyond about 6 days, it is called a paralytic ileus.

C. Causes
   1. Bowel motility is controlled by the autonomic nervous system. Parasympathetic stimulation increases bowel motility, whereas sympathetic stimulation inhibits it.
      a. Increased sympathetic output postoperatively may lead to increased ileus formation. The colon is more dependent on the autonomic nervous system than the stomach or small intestine, which may explain the longer recovery time postoperatively.
      b. The vagal nerve is important to parasympathetic activity in the stomach. Inadvertent damage to the vagal nerve during abdominal surgery can result in impaired emptying of the stomach.

2. During periods of fasting, upper GI tract motility is controlled by the migrating motor complex, which moves intestinal contents distally.
    a. During periods of fasting postoperatively, the contractility of the stomach and small intestine is entirely dependent on the migrating motor complex.
    b. Inflammation of the GI tract after surgery, inhibitory neural reflexes, and the release of inhibitory neurotransmitters such as nitrous oxide, substance P, and vasoactive intestinal peptide may lead to decreased activity of migrating motor complex.
3. Exacerbating factors
    a. Anesthesia: Delayed gastric emptying has been observed with the use of halothane, enflurane, and atropine.
    b. Postoperative opioids: Opioids, through agonism at the mu_2-opioid receptor, slow GI motility mainly by decreasing colonic motility. High doses and prolonged courses postoperatively can contribute to paralytic ileus.
    c. Other medications known to decrease GI motility are often given perioperatively (e.g., anticholinergics).

D. Diagnosis
1. POI is typically characterized by abdominal distension, lack of bowel sounds, delayed passage of flatus or stool, and accumulation of gas and stool in the bowels, which may lead to nausea and vomiting.
2. All patients should have a physical examination for abdominal distension, followed by plain abdominal radiographs to identify air and dilated loops of bowel.
3. An abdominal CT scan can be used to rule out a mechanical obstruction.

E. Management
1. Use of epidural anesthesia
    a. Epidural blockade may improve POI by reducing local sympathetic and inflammatory response postoperatively and increasing splanchnic blood flow. It may also reduce postoperative opioid use.
    b. Several studies of epidural anesthesia have shown reductions in POI. Most studies that showed benefit used thoracic epidural blockade and administered epidural anesthesia for at least 48–72 hours.
2. Use of laparoscopic surgery: Few studies have shown reduced POI rates with laparoscopic surgery compared with open abdominal procedures. However, studies have shown reductions in inflammatory response (e.g., cytokine production) and reduced postoperative pain with laparoscopic procedures, which may affect POI.
3. NG decompression
    a. Historically, NG tubes were placed in most patients for gastric decompression and used until normal GI function returned.
    b. In a meta-analysis of 26 trials including 4000 patients, the use of NG tube insertion was routinely associated with fever, atelectasis, and pneumonia, though patients treated without NG tubes did have more abdominal pain and vomiting. The study concluded that for every patient who required NG tube insertion, 20 patients could be treated effectively without NG tube insertion and that NG tubes should be used selectively because of concerns for adverse effects.
4. Enteral feeding
    a. Traditionally, enteral feeds were delayed postoperatively until after ileus was resolved.
    b. Recent data from several randomized controlled trials show a modest improvement in POI resolution from early enteral feedings, typically initiated within the first 24 hours postoperatively. This effect probably results from stimulation of the bowel.

5. Sham feeding
    a. Sham feeding is the process of eliciting the release of hormonal and neuronal GI activity without regular feeding. One way by which this mechanism can occur is through the chewing of gum.
    b. In a small randomized study of 19 patients, chewing gum three times a day reduced time to first flatus (2.1 days vs. 3.2 days, $p<0.01$) and time to first defecation (3.1 days vs. 5.8 days, $p<0.01$) compared with control.
6. Pharmacologic therapy
    a. Opioid-sparing analgesia
        i. NSAIDs have two potential effects on resolving POI: sparing opioids through their analgesic effects and reducing the production of inflammatory mediators (e.g., prostaglandin). Adding an NSAID to opioid therapy reduces the need for opioids by 20%–30%.
        ii. A randomized, double-blind study of morphine patient-controlled analgesia with or without ketorolac showed decreased morphine use and earlier first bowel movements (1.5 days vs. 1.7 days, $p<0.05$) in the patients who received the additional ketorolac compared with those who did not.
        iii. Use should be carefully assessed so that the benefit outweighs the risk of postoperative bleeding caused by platelet inhibition together with the potential increased risk of anastomotic leak and wound dehiscence.
    b. Prokinetics and laxatives
        i. Erythromycin
            (a) Erythromycin is a macrolide antibiotic that has prokinetic activity as a motilin receptor agonist. Motilin induces gastric contractions and migrating motor complex.
            (b) In randomized controlled trials, erythromycin does not appear beneficial for POI resolution.
        ii. Metoclopramide. No studies of POI have shown a benefit; however, the antiemetic activity of metoclopramide may be beneficial as an adjunctive therapy for patients with POI.
        iii. Laxatives
            (a) Laxatives should play an important role in the management of POI because of their stimulatory action in the GI tract; however, data are limited on the use of laxatives.
            (b) Most of the data regarding the use of laxatives for POI support the use of bisacodyl suppositories (e.g., 10 mg rectally daily), which have reduced time to return of normal bowel function and some evidence of reduced hospital length of stay.
        iv. Gastrografin
            (a) Gastrografin is an oral contrast agent with theoretical benefit in POI because it is hypertonic and reduces gut wall edema by drawing fluid into the gut lumen and promoting peristalsis.
            (b) In a randomized controlled trial of 80 patients with prolonged POI, gastrografin did not significantly reduce the mean duration of POI compared with placebo (83.7 vs. 101.3 hours, $p=0.19$).
    c. Peripherally acting mu-opioid receptor antagonists
        i. Alvimopan (Entereg)
            (a) Alvimopan has 200-fold selectivity for the peripheral opioid receptors and has poor absorption from the GI tract when administered orally (bioavailability about 6%), decreasing the likelihood of systemic absorption and penetration across the blood-brain barrier.

(b) Alvimopan was studied in four North American phase III randomized controlled trials for POI. Each trial included adult surgical patients (generally bowel resection and abdominal hysterectomy) who received standard postoperative care for prevention of POI. In all four trials, patients were randomized to receive placebo, alvimopan 6 mg orally twice daily, or alvimopan 12 mg orally twice daily beginning immediately before surgery and continuing for 7 days (total of 15 doses).
   (1) Three of the four trials showed significant reductions in time to return of normal bowel function (i.e., tolerating solid food and passing bowel movements). The 12-mg dose was especially beneficial in females and patients older than 65.
   (2) A pooled analysis from the four phase III trials comparing outcomes in patients who received the 12-mg dose versus placebo showed a 20-hour reduction in time to return of normal bowel function (102 hours vs. 121.8 hours, $p<0.05$) and a reduction in hospital length of stay (6.6 days vs. 7.6 days, $p<0.001$).

(c) Alvimopan was also associated with significantly less POI-related morbidity, including NG tube insertion (6.6% vs. 11.5%, $p<0.001$), and fewer POI complications, including paralytic ileus, as well as adverse events of nausea, vomiting, and abdominal distension (2.9% vs. 8.8%, $p<0.001$).

(d) In a 12-month study of alvimopan for opioid-induced bowel dysfunction in patients with chronic non-cancer pain, myocardial infarction rates were higher in patients treated with alvimopan than in placebo (7 [1.3%] vs. 0 [0%]).
   (1) This higher risk did not appear to be related to the therapy duration (12 months). In addition, increased risk of cardiovascular events has not occurred in other alvimopan studies, including POI studies, and no causal relationship has been established.
   (2) However, given the concern of added risk, the FDA has since developed REMS (Risk Evaluation and Mitigation Strategies) for the use of alvimopan. Hospitals that intend to use alvimopan should be enrolled in the EASE (ENTEREG Access Support & Education) program. The EASE program limits the use of alvimopan to short-term inpatient use, and patients cannot receive more than 15 doses.

ii. Methylnaltrexone
   (a) Methylnaltrexone is a peripherally acting mu-opioid receptor antagonist approved for treatment of chronic opioid-induced constipation. Methylnaltrexone is not FDA approved for the treatment of POI.
   (b) Methylnaltrexone is typically given subcutaneously daily or every other day for opioid-induced constipation in a weight-based dose of 0.15 mg/kg rounded to the nearest 2 mg (usually 8 or 12 mg)
   (c) Results from phase III trials showed no reduction in duration of POI compared with placebo, but methylnaltrexone may be an alternative to alvimopan if oral therapy is not an option.

iii. Naloxegol
   (a) Naloxegol is another oral peripherally acting mu-opioid receptor antagonist, which is approved for opioid-induced constipation. Dosing for opioid-induced constipation is 25 mg orally once daily, with dosing reductions required for renal dysfunction and concomitant CYP3A4 inhibitors.
   (b) Like methylnaltrexone, naloxegol is not FDA approved for the treatment of POI.

Patient Case

5. A 40-year-old man presents for a large bowel resection. Because of the complexity of the procedure and the anticipated use of perioperative opioids and inhalation anesthesia, the team is creating a plan to prevent POI. Which medication would be best to recommend for reducing the incidence of POI?
   A. Hydromorphone patient-controlled analgesia postoperatively instead of morphine.
   B. Alvimopan 12 mg orally just before surgery and continued twice daily for 7 days.
   C. Metoclopramide 5 mg intravenously every 6 hours for 7 days postoperatively.
   D. Octreotide 100 mcg subcutaneously every 8 hours for 7 days postoperatively.

V. POSTOPERATIVE NAUSEA AND VOMITING

A. Epidemiology
   1. Postoperative vomiting occurs in about 30% of overall surgical patients, whereas postoperative nausea occurs in about 50%. In high-risk patients, the incidence of PONV can be as high as 80%.
   2. Uncontrolled PONV may result in a prolonged stay in the post-anesthesia care unit and, sometimes, unplanned hospital admissions for outpatient procedures.

B. Risk Factors
   1. Surgical factors that increase the risk of PONV:
      a. Use of volatile anesthetics: The effect of volatile anesthetics on PONV is usually dose-dependent and typically presents within the first 6 hours postoperatively.
      b. Use of postoperative opioids: The effect of postoperative opioids similarly increases the risk of PONV in a dose-dependent fashion.
      c. Use of general anesthesia
      d. Duration of anesthesia: Each 1-hour increase in duration of anesthesia increases the risk of PONV.
      e. Type of surgery: Cholecystectomy, laparoscopic, gynecological are most commonly associated with PONV.
   2. Patient-specific factors that increase the risk of PONV:
      a. Female sex
      b. History of motion sickness, chemotherapy-induced nausea and vomiting, or PONV
      c. Nonsmoker
      d. Younger age
   3. The Apfel simplified risk score is based on four predictors: female sex, history of PONV and motion sickness, nonsmoking status, and use of postoperative opioids. The estimated risk of PONV is 10%, 20%, 40%, 60%, and 80% when zero, one, two, three, or four of the above risk factors are present, respectively.
   4. A stratified prevention strategy is suggested according to the number of risk factors, as follows:
      a. Low risk (zero risk factors): No pharmacologic prophylaxis recommended; a "wait and see" strategy is suggested
      b. Medium risk (one or two risk factors): One or two prophylactic interventions are recommended.
      c. High risk (more than two risk factors): Three or more prophylactic interventions should be used.
      d. Of note, new guidelines recommend that at least one pharmacologic prophylactic therapy be administered to all patients regardless of risk factors because risk scores are not completely predictive.

C. Prevention
   1. A multimodal strategy that usually targets risk avoidance should be implemented to avoid PONV. The IMPACT study enrolled more than 5000 surgical patients and evaluated six interventions for the prevention of PONV in a factorial design, including avoidance of volatile anesthetics and nitrous oxide, use of short-acting opioids in the postoperative period, and use of ondansetron, dexamethasone, and droperidol for pharmacologic prophylaxis. Each intervention except for choice of opioid was associated with a reduction in the incidence of PONV; however, the study showed that the interventions acted independently of each other, and thus, a benefit with several interventions was seen.
   2. Regional anesthetics should be used instead of general anesthetics, when possible, because general anesthetics are associated with an 11-fold increase in PONV. If general anesthesia is required, propofol is preferred to volatile anesthetics for inducing and maintaining anesthesia. In the IMPACT study, the use of propofol compared with volatile anesthetics was associated with a 19% relative risk reduction in the incidence of PONV ($p<0.001$).
   3. Avoiding nitrous oxide as a carrier gas is recommended; use of other carrier gases (e.g., nitrogen, oxygen) is associated with a 12% relative risk reduction in PONV ($p=0.003$).
   4. Perioperative opioid use should be minimized, if possible. Guidelines recommend the use of perioperative NSAIDs for opioid-sparing analgesia. Intravenous acetaminophen has also been studied as part of a multimodal analgesia strategy to reduce postoperative opioid use; although oral acetaminophen may also be used, its effect on PONV has not been well studied. According to data from the IMPACT study, which compared remifentanil with fentanyl, the use of short-acting opioids does not appear to be associated with the incidence of PONV.
   5. Pharmacologic prophylaxis:
      a. Serotonin-3 antagonists are first-line treatment for pharmacologic prophylaxis. These drugs are most effective for the prevention of PONV when given at the end of surgery.
         i. Ondansetron, when given at a prophylactic dose of 4 mg intravenously, has greater anti-vomiting effects (number needed to treat [NNT] = 6) than antinausea effects (NNT = 7).
         ii. Granisetron at doses of 0.35–3 mg intravenously is as effective as ondansetron, whereas palonosetron 0.075 mg intravenously was more effective than ondansetron at preventing PONV in a small study of gynecological laparoscopic surgical patients (42% PONV with palonosetron vs. 67% with ondansetron, $p<0.05$).
      b. Dexamethasone is most effective for the prevention of PONV when given at the time of induction.
         i. According to data from the IMPACT study, dexamethasone 4 mg intravenously was associated with a 26% relative risk reduction in the incidence of PONV.
         ii. A recent study evaluating a higher dexamethasone dose (e.g., 8 mg intravenously) at the time of induction for the prevention of PONV found that this high-dose strategy resulted in an 8% absolute risk reduction in vomiting compared with standard PONV care.
      c. Droperidol at prophylactic doses of 0.625–1.25 mg intravenously is effective for the prevention of PONV when given at the end of surgery.
         i. Efficacy of droperidol is similar to that of ondansetron and dexamethasone, with a relative risk reduction of about 25% (data from the IMPACT study).
         ii. FDA black box warnings for cardiovascular risk with droperidol have limited its use; however, the doses used for preventing PONV are very low and are unlikely to be associated with cardiovascular effects.

d. Neurokinin-1 receptor antagonists (e.g., aprepitant) should be given at the time of induction. Data are limited on the use of neurokinin-1 receptor antagonists in PONV; however, they appear to be as effective as ondansetron. One large randomized double-blind trial evaluated 805 abdominal surgery patients who were randomly assigned to receive 40 mg of oral aprepitant, 125 mg of oral aprepitant, or 4 mg of intravenous ondansetron. Although there was no difference in the primary outcome of complete response (considered no vomiting or use of rescue therapy), aprepitant at both doses did reduce the occurrence of vomiting compared with ondansetron.
e. Transdermal scopolamine can be applied the evening before surgery. When used with ondansetron, adding scopolamine was associated with a 10% absolute reduction in the occurrence of PONV within the first 24 hours postoperatively.

D. Rescue Therapy
1. When rescue therapy is needed within the first 6 hours postoperatively, an antiemetic should be selected from a therapeutic class different from the initial prophylactic drug. Repeat doses of the same drugs that were used for initial prophylaxis can be tried if PONV occurs more than 6 hours after surgery.
2. If patients did not receive a prophylactic agent, rescue treatment with a serotonin-3 antagonist should be tried.

Patient Case

6. A 27-year-old woman presents for a total abdominal hysterectomy. She is a nonsmoker who has a significant history of motion sickness. During the procedure, she is expected to receive general anesthesia with volatile anesthetics. She will probably require high-dose opioids perioperatively. Given this patient's risk of developing PONV, which would she best receive for prevention of PONV?
   A. Patient has a high risk of PONV; she should receive transdermal scopolamine the evening before surgery, dexamethasone 4 mg intravenously at the time of induction, and ondansetron 4 mg intravenously at the end of surgery.
   B. Patient has a moderate risk of PONV; she should receive dexamethasone 4 mg intravenously at the time of induction and ondansetron 4 mg intravenously at the end of surgery.
   C. Patient has a high risk of PONV; she should receive dexamethasone 4 mg intravenously at the time of induction and ondansetron 4 mg intravenously at the end of surgery.
   D. Patient has a low risk of PONV; she should receive ondansetron 4 mg intravenously at the end of surgery.

VI. UPPER GASTROINTESTINAL BLEEDING

A. Definition and Epidemiology
1. Bleeding that occurs in the esophagus, stomach, or duodenum
2. In the United States, the annual incidence is 65 hospitalizations per 100,000 adults. Incidence is twice as high in males as in females and increases with age.
3. Accounts for 350,000 hospitalizations per year at an annual cost of $2.5 billion
4. UGIB is 4 times more common than lower GI bleeding, and the hospitalization rate is around 6-fold higher.

5. Mortality at 28 days after hospital admission for non-variceal hemorrhage is about 13%, whereas mortality after variceal hemorrhage is about 20%. Mortality rates after both UGIB classifications seem to be decreasing with advances in care.

B. Etiologies
   1. Nonvariceal hemorrhage
      a. Gastric and/or duodenal ulcer
         i. Most common cause of severe cases of UGIB, accounting for 47% of all UGIB episodes
         ii. Bleeding is more common in the setting of anticoagulant use and is typically self-limited.
         iii. Most commonly caused by an *H. pylori* infection, but may also be secondary to NSAIDs, hyperacidity (e.g., in Zollinger-Ellison syndrome), or stress-related mucosal disease (stress-related mucosal damage is discussed in detail in the Supportive and Preventive Medicine chapter)
      b. Esophagitis
         i. Noted in around 13% of patients with UGIB
         ii. Risk factors include history of gastroesophageal reflux disease, medication use (e.g., NSAIDs, oral bisphosphonates, tetracycline), and infection (e.g., *Candida*, herpes simplex virus).
         iii. Commonly presents with hematemesis and less commonly associated with melena
         iv. Lower rebleeding and mortality rate than other sources of UGIB
      c. Gastroduodenal erosions (erosive gastritis/duodenitis): Defects of the gastric/duodenal mucosal layer that lead to inflammation without ulcer formation
         i. Causes are similar to those for gastric/duodenal ulcers. Additional risk factors include excessive alcohol consumption, radiation injury, bariatric surgery, and chronic bile reflux.
         ii. Bleeding is more common in the setting of anticoagulant use and is typically self-limited.
         iii. May progress to ulcer formation
      d. Mallory-Weiss tear: A longitudinal mucosal laceration in the distal esophagus and proximal stomach (an intramural dissection)
         i. A sudden increase in intra-abdominal pressure leads to forceful distention of the gastroesophageal junction and a resultant mucosal tear.
         ii. Typically caused by forceful vomiting
      e. Less common causes: Vascular malformation, malignant formations, aortoenteric fistulas, gastric antral vascular ectasia, and prolapse gastropathy
   2. Variceal hemorrhage
      a. Relatively few severe cases of UGIB are secondary to variceal hemorrhage (< 5%), but the incidence is higher in patients with cirrhosis.
      b. Mortality rate is around 20% at 6 weeks.
      c. Portal hypertension caused by the obstruction of venous blood flow through the cirrhotic liver leads to increased pressure in the portal vein and causes the redirection of blood flow to other areas of the body.
      d. Varices may be present in any portion of the GI tract, but they are most common in the esophagus and stomach.
      e. Gastroesophageal varices are present in about 50% of patients with cirrhosis. About 12% of patients with varices will develop variceal hemorrhage within 1 year of diagnosis, and the recurrence rate for variceal hemorrhage within 1 year is about 60%.

C. Initial Assessment and Risk Stratification
   1. Most patients (50%) present with both melena and hematemesis; 30% have hematemesis (either red blood or "coffee-ground" emesis) alone, and 20% have melena alone.
   2. Hematochezia (bright red blood per rectum) may also be present (in about 5% of patients), which may represent a swift UGIB (about 1 L of blood is needed in the stomach to cause hematochezia, whereas only 50–100 mL is needed to cause melena).

3. Bleeding ulcers may result in right upper quadrant pain. Mallory-Weiss tears may present as emesis, retching, or coughing before hematemesis. Patients with symptoms associated with chronic liver disease will likely have variceal bleeding.
4. Hemodynamic instability may be present in patients with significant hypovolemia. Initial care of these patients should focus on patient stabilization.
5. Insertion of an NG tube is controversial because it has not been shown to improve clinical outcomes, but inspection of the aspirate may be useful in patients without frank hematemesis.
    a. If the aspirate contains bright red blood, urgent endoscopy is likely indicated.
    b. A normal-appearing aspirate does not rule out UGIB because about 15% of patients with a normal aspirate have high-risk lesions on endoscopy.
    c. Insertion of an NG tube may be contraindicated in patients with a history of varices, particularly those with recent endoscopic band ligation.
6. In patients with a variceal hemorrhage, a hepatic venous pressure gradient greater than 20 mm Hg is a strong predictor of early rebleeding and death and can be used for risk stratification. Measuring this gradient is not feasible at most centers, but more than 80% of patients with Child-Turcotte-Pugh class C have a gradient greater than 20 mm Hg.
7. Scoring tools may help in patient risk stratification, which can aid in site of care and endoscopy timing decisions.
    a. The Blatchford scoring system uses clinical and laboratory parameters to predict the need for clinical intervention.
    b. The Rockall scoring system incorporates endoscopic findings and predicts a patient's risk of rebleeding and death.
    c. The AIMS65 scoring system uses clinical and laboratory parameters to estimate in-hospital mortality, hospital length of stay, and cost in patients with acute UGIB.
    d. Current guidelines recommend using a Blatchford score of 1 or less to identify patients at very low risk of rebleeding or mortality who may not require hospitalization or inpatient endoscopy. The guidelines do not make a recommendation for or against using the Rockall score and recommend against using the AIMS65 scoring system to identify patients at low risk of rebleeding or mortality.

D. Management
    1. General measures
        a. Venous access with two large-caliber (at least 18 gauge; 16 gauge preferred if hemodynamically unstable) peripheral intravenous catheters should be achieved. Access by peripheral intravenous catheters is preferred to central venous catheterization because of their improved ability to deliver intravenous fluids more quickly (because of the Poiseuille law).
        b. Supplemental oxygen by nasal cannula should be administered to patients with an oxygen saturation below 90%.
        c. A blood type and cross-match should be sent immediately (in preparation for possible blood transfusion).
    2. Hemodynamically unstable patients should be resuscitated immediately with intravenous fluids (crystalloids) and blood transfusions (if indicated).
        a. A study of patients with hemodynamic instability secondary to UGIB compared usual care with intensive resuscitation focused on achieving hemodynamic stability, a hematocrit greater than 28%, and an INR less than 1.8. Intensive resuscitation was associated with a lower mortality rate (2.8% vs. 11.1%, p=0.04) and a lower incidence of myocardial infarction (5.6% vs. 13.9%, p=0.04).
        b. See the Shock Syndromes II: Hypovolemic, Critical Bleeding, and Obstructive chapter for further discussion of hypovolemic shock secondary to hemorrhage.

3. Transfusion
    a. Current guidelines recommend a transfusion threshold of 7 g/dL for hospitalized hemodynamically stable patients, including those in the ICU, and a threshold of 8 g/dL in patients with preexisting cardiovascular disease. This recommendation does not extend to patients with acute coronary syndrome.
    b. A higher hemoglobin threshold may be appropriate in patients with hypotension or who are bleeding rapidly.
    c. Platelet transfusion, typically if the platelet count is less than 50,000/mm^3, should be considered.
4. Endoscopy
    a. Diagnostic endoscopy
        i. Used to diagnose and assess the risk posed by the bleeding lesion(s). Therapeutic endoscopy may also be used for the lesion(s) to reduce the risk of bleeding recurrence (discussed later in the chapter).
        ii. Patients with UGIB should generally have a diagnostic endoscopy within 24 hours.
        iii. Patients who are hemodynamically unstable and those with a suspected variceal UGIB should have a diagnostic endoscopy as soon as possible and no later than 12 hours after presentation.
        iv. Erythromycin or metoclopramide may be used when a large amount of blood in the stomach would hinder an endoscopy; however, routine use is not recommended. Use reduces the need for repeated endoscopy (OR 0.55; 95% CI, 0.32–0.94) but does not alter the need for blood products, length of hospital stay, or need for surgery.
        v. Endotracheal intubation before endoscopy may be indicated to prevent aspiration, but patient selection is controversial.
        vi. Endoscopic findings predict the risk of rebleeding and guide further therapies.
            (a) Nonvariceal UGIB: Stigmata of recent hemorrhage from a peptic ulcer predict the risk of further bleeding and guide management decisions (Table 6).

Table 6. Endoscopic Findings of Bleeding Peptic Ulcers, Prevalence, and Rebleeding Rate

Risk of Rebleeding	Stigmata of Recent Hemorrhage	Forrest Grade	Prevalence	Rebleeding Rate
High	Active spurting bleeding	IA	9.3% (spurting and oozing)	55% (spurting and oozing)
	Active oozing bleeding	IB		
	Nonbleeding visible vessel	IIA	6.1%	43%
	Adherent clot	IIB	6.5%	22%
Low	Flat pigmented spot	IIC	13.1%	10%
	Clean base	III	52.6%	5%

Information from: Laine L, Peterson W. Bleeding peptic ulcer. N Engl J Med 1994;331:717-27; Enestvedt BK, Gralnek IM, Mattek N, et al. An evaluation of endoscopic indication and findings related to nonvariceal upper-GI hemorrhage in a large multicenter consortium. Gastrointest Endosc 2008;67:422-9; and Laine L, Jensen DM. Management of patients with ulcer bleeding. Am J Gastroenterol 2012;107:345-60.

            (b) Variceal UGIB: If active variceal hemorrhage is confirmed, endoscopic and pharmacologic therapy should be initiated.
    b. Therapeutic endoscopy
        i. Can be used in conjunction with a diagnostic endoscopy to treat the source of UGIB once it has been identified
        ii. Non-variceal UGIB
            (a) Peptic ulcer
                (1) Endoscopic therapy should be used for lesions with active spurting, active oozing, or a non-bleeding visible vessel.

(2) Endoscopic therapy for an adherent clot resistant to vigorous irrigation is controversial. Patients with clinical features associated with a higher risk of rebleeding (e.g., older age) may benefit from endoscopic therapy.
(3) Endoscopic therapy should not be used for ulcers with a flat pigmented spot or clean base.
(4) Endoscopic therapy includes hemostatic clips, epinephrine injection, and thermal therapy. Often, these therapies are combined. Epinephrine injection should not be used alone but should be combined with a second modality.

(b) Other findings (e.g., erosions or Mallory-Weiss tear) are typically unamenable to endoscopic therapy.

iii. Variceal UGIB
(a) The primary endoscopic therapy for esophageal varices is endoscopic band ligation, which is preferred to sclerotherapy.
(b) Sclerotherapy is recommended in patients for whom endoscopic band ligation is unfeasible.
(c) Gastric varices are often unamenable to endoscopic band ligation, and rescue therapies may be necessary (discussed later in the chapter).

5. Pharmacologic management
a. Non-variceal UGIB
i. Acid-suppressive therapy is the mainstay of pharmacologic therapy for acute non-variceal UGIB.
ii. Gastric acid inhibits platelet aggregation, impairs clot formation, and promotes fibrinolysis. In in vitro studies, inhibiting gastric acid and raising the intragastric pH to 6 or higher may promote clot formation and decrease the risk of rebleeding. However, clinically, a pH of 4–5 should likely be sufficient.
iii. Histamine-2 receptor antagonists are ineffective at achieving a pH greater than 6 and have shown inconsistent benefit; therefore, these agents are not recommended for patients with acute ulcer-related UGIB.
iv. Adding octreotide to PPI therapy does not appear to reduce hospital or ICU length of stay, rebleeding, or mortality in patients with nonvariceal UGIB and is not routinely recommended.
v. High-dose PPIs should be used as an adjunct to endoscopic therapy.
(a) The most recent American College of Gastroenterology guidelines do not recommend for or against the use of pre-endoscopic PPIs in patients with UGIB. This is a change from prior versions, which stated that pre-endoscopic PPIs could be considered to decrease the need for endoscopic therapy.
(b) High-dose PPI is defined as 80 mg/day or more for 3 days or more. This can be administered as a continuous infusion or as intermittent boluses.
(c) High-risk patients with UGIB because of ulcers who underwent endoscopic therapy should receive high-dose PPI treatment for 3 days, followed by twice-daily therapy for 2 weeks after the index endoscopy.

b. Variceal UGIB
i. Octreotide, a somatostatin analog, should be initiated promptly when variceal UGIB is suspected and continued for 2–5 days after the diagnosis is confirmed.
(a) Octreotide causes selective splanchnic vasoconstriction through inhibition of the release of vasodilatory peptides (mainly glucagon). Octreotide may also have a local vasoconstrictive effect. Splanchnic vasoconstriction decreases portal inflow, which indirectly reduces variceal blood flow and hemorrhage volume.
(b) Typically given as a 50-mcg bolus, followed by a 50-mcg/hour continuous intravenous infusion

(c) When used without therapeutic endoscopy, octreotide is only marginally beneficial (i.e., reduction of packed red blood transfusion by 0.7 unit with no benefit on rebleeding or mortality).
(d) Compared with endoscopic therapy alone, a somatostatin analog combined with endoscopic therapy is associated with improved initial control of bleeding (relative risk [RR] 1.12; 95% CI, 1.02–1.23) and hemostasis at 5 days (RR 1.28; 95% CI, 1.18–1.39) with no difference in mortality or serious adverse events.
(e) Patients should be monitored for bradycardia and hyperglycemia during octreotide infusion.
(f) For patients in whom a TIPS is performed successfully, octreotide can be discontinued.

ii. Vasopressin infusion is not recommended for variceal UGIB because of the high incidence of adverse events (cardiac, peripheral, and bowel ischemia) with doses necessary to reduce splanchnic blood flow (0.2–0.8 unit/minute).
iii. Because of the high incidence of peptic ulcer–related UGIB, high-dose PPI should be initiated, even when variceal UGIB is suspected, until the diagnosis of variceal UGIB is confirmed.
(a) There is no evidence that high-dose PPI therapy reduces the risk of rebleeding after endoscopic therapy for variceal UGIB.
iv. Patients with cirrhosis, with or without ascites, and UGIB (whether variceal or non-variceal) should be initiated on short-term prophylactic antibiotics.
(a) Antibiotics are associated with a lower risk of infection, lower risk of rebleeding, shorter length of stay, and higher survival rates.
(b) Guidelines recommend therapy with ceftriaxone 1 g daily. Ceftriaxone is preferred to fluoroquinolones because of the high prevalence of fluoroquinolone resistance. Ceftriaxone should be considered for discontinuation when hemorrhage has resolved and octreotide is discontinued.
(c) Prophylactic antibiotic therapy should be continued for no more than 7 days.
v. For patients in whom a TIPS is not performed, a nonselective β-blocker (propranolol, nadolol, or carvedilol) should be initiated once octreotide is discontinued unless the patient's heart rate or blood pressure prohibit β-blocker therapy.
vi. Simvastatin may be added to standard therapy in patients with cirrhosis and variceal bleeding.
(a) A randomized placebo-controlled trial showed that simvastatin did not reduce the rate of rebleeding but was associated with decreased mortality in patients with Child-Pugh class A and B cirrhosis.
(b) Patients should be monitored closely for clinical signs and symptoms of rhabdomyolysis.

c. Treatment of *H. pylori* infection is beyond the scope of this chapter; however, a 14-day treatment should be given to all patients with suspected or diagnosed infection, and eradication should be confirmed 4 weeks after therapy.

6. Rescue therapies
a. For patients in whom endoscopic therapy has failed or who are not candidates for endoscopy, angiographic intervention (typically selective arterial embolization) may be required.
b. For variceal UGIB, balloon tamponade may be used as a temporizing method (maximum 24 hours) while definitive therapy is planned.
c. TIPS is indicated in patients in whom hemorrhage from esophageal varices cannot be controlled or in whom bleeding recurs, despite pharmacologic and endoscopic therapy. Patients with Child-Turcotte-Pugh class C cirrhosis or Child-Turcotte-Pugh class B with bleeding on endoscopy may be considered for TIPS within 72 hours.

Patient Case

7. A 27-year-old man with a medical history of Crohn disease presents to your ED with frank bloody output from his rectum. The patient has hypotension (systolic blood pressure 85 mm Hg, MAP 58 mm Hg), and an NG lavage reveals "coffee-ground" material. The resident on call is in the process of calling the endoscopy team to help diagnose and intervene on an UGIB with an esophagogastroduodenoscopy. Which statement is most accurate regarding the care of this patient?
   A. An endoscopy is the appropriate course of action but should not be performed until after initiation of PPI continuous infusion.
   B. A PPI continuous infusion should be initiated as soon as possible and continued for 48 hours post-endoscopy.
   C. The patient, who likely has a lower GI bleed, would benefit from a colonoscopy as opposed to an esophagogastroduodenoscopy.
   D. An endoscopy and a PPI continuous infusion should be initiated as soon as possible.

E. Management of Anticoagulation and Antiplatelet Therapy
   1. Anticoagulation
      a. Patients who do not have a strong indication for continued anticoagulation at the time of the bleeding event should have antithrombotic therapy discontinued.
      b. American College of Gastroenterology recommends against the use of fresh frozen plasma or vitamin K in patients on warfarin hospitalized with GI bleed. Although this group could not reach a consensus regarding the role of PCC for management of GI bleeding in patients receiving warfarin, guidelines suggest the use of PCC over FFP, particularly when bleeding is life-threatening, INR is significantly elevated, or massive blood transfusion is not feasible.
      c. Reversal of dabigatran with idarucizumab is not routinely recommended but may be considered in patients with a life-threatening GI bleed who have taken dabigatran in the past 24 hours.
      d. Use of andexanet alfa is not recommended for patients taking rivaroxaban or apixaban who are hospitalized with a GI bleed.
      e. Administration of PCC for patients hospitalized with GI bleed who have been taking any direct oral anticoagulant is not routinely recommended. However, use may be considered in patients who have taken any of these agents in the past 24 hours and have a life-threatening bleed.
      f. Most patients who develop UGIB while receiving long-term antithrombotic therapy continue to be at risk of thrombosis; these patients should be resumed on antithrombotic therapy.
         i. In a retrospective cohort study of patients who had a GI hemorrhage during warfarin therapy, those who resumed warfarin within 90 days after the hemorrhagic event had a lower adjusted risk of death (HR 0.31; 95% CI, 0.15–0.62) and thrombosis (HR 0.05; 95% CI, 0.01–0.58), without a significant increase in the risk of recurrent GI hemorrhage (HR 1.32; 95% CI, 0.50–3.57). In this study, warfarin was resumed a median of 4 days (interquartile range 2, 9) after presentation for UGIB. These data suggest that for many patients with warfarin-associated GI hemorrhage, the benefits of resuming warfarin therapy outweigh the risks.
         ii. Antithrombotic therapy should be withheld at minimum until the source of bleeding is found and controlled.
         iii. The patient's short-term risk of rebleeding should be weighed against the short-term risk of thrombosis when deciding to resume antithrombotic therapy.
         iv. Some experts recommend that warfarin be resumed 4–7 days after presentation for UGIB.
         v. The risks, benefits, and timing of resuming target-specific oral anticoagulants are unclear.

2. Antiplatelet therapy
    a. For patients on aspirin for secondary cardiovascular prevention who present with GI bleed, holding aspirin therapy is not recommended. If aspirin is interrupted, it should be resumed within 24 hours of achieving hemostasis.
    b. Timing of reinitiation of other antiplatelet therapy depends on the risk of rebleed determined during endoscopy. If this risk is considered low, P2Y12 receptor blockers may be restarted within 1–3 days after hemostasis. Reinitiation of these agents is further delayed in high-risk patients until the risk has decreased.

VII. ENDOCRINE EMERGENCIES

A. Epidemiology
1. Diabetic ketoacidosis (DKA) and hyperosmolar hyperglycemic state (HHS) are the most common diabetic emergencies.
    a. DKA accounts for 500,000 hospital days and costs up to $5 billion per year in the United States.
    b. Between 2009 and 2014, there was a 54% increase in hospitalization rates for DKA in the United States, with the highest rates in patients younger than 44.
    c. Most patients who develop DKA have type 1 diabetes, and DKA is considered the most important contributor to mortality rates in children and adolescents with diabetes.
    d. In-hospital mortality rates for DKA are 0.1%–2.6%, and can be as high as 5%–20% for HHS, depending on age and other comorbidities.
2. Hyperglycemia (BG above 140 mg/dL) occurs commonly in the setting of critical illness, with prevalence rates depending on the level of hyperglycemia and patient population evaluated.
    a. About 27% of critically ill patients have a BG above 200 mg/dL on ICU admission, and about 90% of patients will have at least one BG reading above 110 mg/dL during their ICU stay.
    b. Hyperglycemia has consistently been associated with increased mortality in critically ill patients, most notably in patients without diabetes. The link between hyperglycemia and worse outcomes seems strongest for patients with acute coronary syndrome or stroke.
    c. Glucose variability (BG fluctuation over time) has also been associated with ICU mortality.
3. Hypoglycemia also occurs commonly in the ICU, both with and without intensive glucose control.
    a. In a large international cohort of patients, moderate hypoglycemia (BG less than 70 mg/dL) occurred in 37% of patients and was independently associated with mortality on multivariable regression (OR 1.78; 95% CI, 1.39–2.27).
    b. In a secondary analysis of the Normoglycemia in Intensive Care Evaluation–Survival Using Glucose Algorithm Regulation (NICE-SUGAR) study, moderate hypoglycemia occurred in 45% of patients (74.2% in the intensive-control group and 15.8% in the conventional-control group), and severe hypoglycemia (BG below 40 mg/dL) occurred in 3.7% of patients (6.9% in the intensive-control group and 0.5% in the conventional-control group). Both moderate and severe hypoglycemia (HR 1.41; 95% CI, 1.21–1.62 and HR 2.10; 95% CI, 1.59–2.77, respectively) were independently associated with death compared with the absence of hypoglycemia.
4. Thyroid crisis, also known as thyroid storm or critical thyrotoxicosis, is an uncommon manifestation of hyperthyroidism known to occur in less than 10% of patients admitted for thyrotoxicosis. Thyrotoxicosis has been associated with mortality rates of 8%–25%, if treated, and up to 100%, if untreated.
5. Myxedema coma is an uncommon severe manifestation of hypothyroidism, with mortality rates of about 20%–25% with appropriate treatment and up to 80% without treatment.

6. Adrenal crisis (i.e., acute adrenal insufficiency or Addisonian crisis) is a life-threatening disorder caused by glucocorticoid (and possibly mineralocorticoid) deficiency.
    a. Adrenal crisis necessitating hospital admission has an incidence of 3.3 per 100 patient-years in those with chronic adrenal insufficiency.
    b. Critical illness–related corticosteroid insufficiency (CIRCI) is a separate entity from adrenal crisis. The prevalence of CIRCI is about 10%–20% but depends on the definition used and the population evaluated. Prevalence rates as high as 60% for patients with septic shock have been reported.

B. DKA and HHS
    1. Clinical presentation of DKA
        a. Current guidelines define DKA as a combination of hyperglycemia (BG greater than 250 mg/dL), ketosis (positive urine or serum ketones), and acidosis (pH less than 7.30) with a serum bicarbonate less than 18 mmol/L and anion gap greater than 10. Patients with severe DKA present with severe acidemia (pH less than 7), a serum bicarbonate less than 10 mmol/L, and an anion gap greater than 12 together with depressed mental status (stupor or coma).
        b. Serum glucose concentrations are usually less than 800 mg/dL but may be more elevated in patients who are comatose. In the setting of starvation, pregnancy, treatment with insulin prior to presentation, or use of sodium-glucose- cotransporter 2 inhibitors, glucose may be normal or only mildly elevated.
        c. A lack of insulin leads to reduced glucose uptake, which, together with increased counterregulatory hormone (catecholamines, cortisol, glucagon, and growth hormone) release, leads to increased lipolysis. The released free fatty acids are metabolized in the liver to ketone bodies (β-hydroxybutyrate and acetoacetate), resulting in ketonemia and elevated anion gap metabolic acidosis.
        d. Although symptoms may exist for a few days, ketoacidosis occurs quickly, and patients may deteriorate rapidly.
        e. Typically occurs in young, leaner patients with type 1 diabetes
        f. Patients may present with symptoms of nausea and vomiting (80% of patients; more indicative of DKA than of HHS), polyuria, polydipsia, weight loss, abdominal pain (30% of patients), and fruity breath from acetone in the blood. Signs of DKA include tachycardia, poor skin turgor, hypotension (because of intravascular volume loss from osmotic diuresis), and Kussmaul respirations (due to severe acidemia).
        g. Measured serum potassium may be high because hyperosmolality and lack of endogenous insulin cause intracellular potassium ions to shift into the extracellular space. However, patients typically have a relative deficiency of total body potassium, which can be worsened with insulin treatment.
    2. Clinical presentation of HHS
        a. HHS is similar to DKA and is defined as BG greater than 600 mg/dL and serum osmolality greater than 320 mOsm/L with pH greater than 7.3 and serum bicarbonate greater than 18 mmol/L. Typically, ketonemia or ketonuria is not present (if present, only in small amounts).
        b. HHS presentation is often later in the disease course than DKA (because of the lack of ketosis), and symptoms evolve over days to weeks. Patients also have higher degrees of dehydration owing to osmotic diuresis.
        c. Typically occurs in older patients with obesity with type 2 diabetes
        d. Signs and symptoms, including electrolyte abnormalities, are similar to those in DKA. However, confusion is much more apparent in HHS and is directly related to the serum osmolality.
        e. Patients with HHS do not normally develop ketoacidosis. Although there is an overall insulin deficiency in both DKA and HHS, there is enough insulin secretion to prevent ketogenesis in patients with HHS.
        f. Lack of access to water, either because of the illness itself or because of an altered thirst response in older adult patients, can worsen the severity of dehydration in the setting of hyperglycemia.

3. Typically, DKA and HHS are caused by an insufficiency of insulin in patients with diabetes combined with another potential trigger such as infection, medication nonadherence, myocardial infarction, cerebrovascular accident, pancreatitis, and certain drugs (i.e., steroids, diuretics, vasopressors, antipsychotics, cocaine, sodium-glucose cotransporter 2 inhibitors).
4. Management
    a. Typically involves fluid resuscitation, correction of hyperglycemia, and electrolyte (mainly potassium) replacement. Correction of acidemia may also be indicated.
    b. Patients are often profoundly hypovolemic. Total body water deficits may be as high as 10–12 L and should be replaced within the first 24 hours.
        i. In the absence of concomitant cardiogenic shock, 0.9% sodium chloride should be administered at a rate of 15–20 mL/kg (typically 1–1.5 L) for the first hour. Thereafter, fluid administration is titrated to hemodynamic parameters and urine output (i.e., fluid infusion rates typically 250–500 mL/hour).
        ii. Patients with mild dehydration and normal or high sodium concentrations can be changed to 0.45% sodium chloride infused at a rate of 250–500 mL/hour.
        iii. Maintenance fluids can be switched to a dextrose-containing fluid (often 5% dextrose with 0.45% sodium chloride) once BG concentrations have dropped to less than 200 mg/dL in DKA and less than 300 mg/dL in HHS.
        iv. Serum sodium concentration should be considered when choosing resuscitation fluid because rapid correction of hypernatremia may have severe consequences.
        v. Recent studies suggest that time to DKA resolution may be shorter when balanced crystalloids are used for resuscitation in place of saline, but further studies are needed to confirm benefit.
    c. Insulin therapy is the main treatment modality of DKA and HHS.
        i. Insulin corrects hyperglycemia and inhibits the release of free fatty acids, which decreases ketone formation and corrects acidosis.
        ii. Intravenous regular insulin is preferred to subcutaneous insulin because of its short half-life and ease of titration.
        iii. Initiate with a 0.1-unit/kg intravenous bolus, followed by a 0.1-unit/kg/hour continuous infusion. Bolus dosing may vary by patient and institution; if bolus is given, BG should be monitored closely to avoid hypoglycemia. An alternative approach, without an initial bolus but initiated at a higher continuous infusion rate (0.14 unit/kg/hour), provides a time similar to DKA resolution.
        iv. The insulin infusion should subsequently be titrated on an hourly basis to decrease BG concentrations by 50–75 mg/dL/hour.
        v. DKA: Decrease dose to 0.02–0.05 unit/kg/hour once BG concentrations drop to 200 mg/dL, and maintain a BG of 150–200 mg/dL until ketoacidosis has resolved.
        vi. HHS: Decrease dose to 0.02–0.05 unit/kg/hour once BG concentrations drop to 300 mg/dL, and maintain a BG of 200–300 mg/dL until mental status changes have resolved.
        vii. Continuous insulin infusions should be continued until ketoacidosis resolves in DKA and abnormal serum osmolality and mental status in HHS resolve; then, change to subcutaneous insulin.
            (a) Criteria for resolution of ketoacidosis include a BG less than 200 mg/dL and two of the following: a serum bicarbonate concentration of 15 mEq/L or greater, a venous pH greater than 7.3, and a normalization of anion gap. Per guidelines, anion gap should be 12 mEq/L or less, but normal range may vary by institution.

(b) For patients with new insulin requirement, doses of 0.5–0.8 units/kg of subcutaneous insulin per day can be used with a basal (glargine or detemir) and bolus (lispro, aspart, or glulisine). For patients previously on subcutaneous insulin therapy, outpatient insulin requirement can be used to guide dosing; weight-based dosing may be insufficient for those who have high baseline requirements. There should be an overlap of 1–2 hours between administration of subcutaneous insulin and discontinuation of intravenous insulin infusion.

(c) Subcutaneous insulin can be used in patients with mild or moderate DKA or HHS but have not been evaluated in patients with severe disease.

(1) Basal insulin is started at a dose of 0.15–0.3 units/kg and repeated every 12–24 hours.

(2) An initial bolus of 0.2 units/kg of rapid acting insulin is also given. Subsequent doses of rapid acting insulin are repeated every 2–4 hours until DKA has resolved.

d. Potassium should be replaced aggressively (assuming adequate renal function) with a goal serum potassium of 4–5 mEq/L.

i. Patients with initial serum potassium less than 3.3 mEq/L should have insulin withheld and potassium administered until their serum potassium is above 3.3 mEq/L.

ii. Patients with serum potassium of 3.3–5.2 mEq/L should have 20–30 mEq of potassium added to each liter of intravenous fluid administered.

iii. Patients with initial serum potassium greater than 5.2 mEq/L should not receive potassium, but they should be monitored closely (with potassium administered if the patient's serum falls below 4 mEq/L). Of importance, patients with DKA and initial hyperkalemia should not receive therapy directed toward total body potassium removal (e.g., sodium polystyrene sulfonate or furosemide). The patient's serum potassium will typically decrease as insulin is administered.

e. Intravenous sodium bicarbonate should not be administered routinely.

i. Acidemia will typically quickly correct as ketosis is resolved, and bicarbonate may have deleterious effects (e.g., hypokalemia).

ii. Studies of patients with DKA with a pH of 6.9 or greater have not supported a benefit with sodium bicarbonate administration.

iii. A recent study of sodium bicarbonate in ICU patients showed a potential mortality benefit in patients with acute kidney injury. However, this study excluded patients with ketoacidosis.

iv. If a patient's pH is below 6.9, sodium bicarbonate should be administered.

(a) Prospective studies of this patient population have not been reported, but this level of acidemia may lead to severe sequelae.

(b) Guidelines recommend that 100 mEq of sodium bicarbonate be placed in 400 mL of sterile water with 20 mEq of potassium chloride, administered at a rate of 200 mL/hour. The infusion should be discontinued if the patient's pH is above 7.

f. Phosphorus repletion should be initiated in patients with serum phosphate below 1 mg/dL or in those with severe cardiac or respiratory dysfunction associated with hypophosphatemia.

C. Hyperglycemia

1. Clinical presentation

a. Because hyperglycemia is so prevalent in critically ill patients, there is no specific clinical presentation.

b. Critically ill patients often do not have the "classic triad" of diabetes symptoms of polyuria, polydipsia, and polyphagia.

c. Stress-induced hyperglycemia has been associated with illness severity.

2. Causes
    a. Critically ill patients have increased release of "stress hormones" (e.g., cortisol and epinephrine) and cytokines. These responses lead to both increased glucose production and insulin resistance, which results in hyperglycemia.
    b. Hyperglycemia is further exacerbated by infusions of dextrose-containing fluids, corticosteroids, and exogenous sympathomimetic medication administration.
3. Management
    a. Hyperglycemia was once considered a beneficial adaptive response in the critically ill population and was not considered a treatment priority. In general, BG was only treated if it exceeded 200 mg/dL.
    b. Tighter glucose control garnered increased interest because hyperglycemia has independently been associated with increased ICU mortality.
    c. A treatment paradigm shift occurred in 2001 with the publication of the landmark "intensive insulin therapy" study.
        i. In this single-center study, surgical ICU patients who were receiving parenteral nutrition were randomized to intensive intravenous insulin therapy (goal BG 80–110 mg/dL) or conventional insulin therapy (goal BG 180–200 mg/dL).
        ii. Patients randomized to intensive insulin therapy had a significantly lower ICU mortality rate than patients receiving conventional insulin therapy (4.6% vs. 8.0%, $p<0.04$). The ICU mortality benefit was most pronounced in patients who stayed in the ICU (and received intensive insulin therapy) for greater than 5 days (10.6% vs. 20.2%, $p=0.005$).
        iii. Patients randomized to intensive insulin therapy also less commonly developed bloodstream infections, acute kidney injury, and ICU-acquired weakness (at the time termed *critical-illness polyneuropathy*).
        iv. A study of medical ICU patients with an identical design from the same center was published in 2006. The study detected no in-hospital mortality benefit with intensive insulin therapy (37.3% vs. 40.0% with conventional insulin therapy, $p=0.33$). Patients in the intensive insulin therapy arm less commonly developed acute kidney injury, had a shorter time to liberation from mechanical ventilation, and had earlier discharge from the ICU and hospital. For patients who stayed in the ICU for 3 days or more, in-hospital mortality was lower in the intensive insulin therapy arm (43.0% vs. 52.5%, $p=0.009$), but the validity of this subgroup analysis has been called into question because patients were not defined by a baseline characteristic.
        v. In light of these findings, intensive insulin therapy was widely recommended by treatment guidelines (including the 2008 Surviving Sepsis Campaign guidelines) and often implemented into practice as a standard of care.
    d. Multicenter trials do not support routine use of intensive insulin therapy.
        i. Because of concern with the single-center nature of the aforementioned studies, unblinded design, and large relative mortality benefit, three multicenter trials were designed and conducted.
        ii. One multicenter study was terminated early because of safety concerns. Patients randomized to intensive insulin therapy more commonly developed severe hypoglycemia (BG of 40 mg/dL or less) than those allocated to conventional insulin therapy (17.0% vs. 4.1%, $p<0.001$). Intensive insulin therapy was not associated with a benefit in 28-day mortality (24.7% vs. 26.0%, $p=0.74$), but the study was inadequately powered to assess this outcome.
        iii. A second multicenter study was terminated early because of several protocol violations. Intensive insulin therapy was not associated with a benefit in ICU mortality (17.2% vs. 15.3%, $p=0.41$), but patients allocated to intensive insulin therapy more commonly developed severe hypoglycemia (BG of 40 mg/dL or less; 8.7% vs. 2.7%, $p<0.0001$).

iv. The NICE-SUGAR study, evaluated intensive insulin therapy (target BG 81–108 mg/dL) or conventional glucose control (BG of 180 mg/dL or less) in more than 3000 patients.
(a) At 90 days, intensive insulin therapy was associated with increased mortality (27.5% vs. 24.9%, p=0.02).
(b) Severe hypoglycemia (BG of 40 mg/dL or less) was more common in patients allocated to intensive insulin therapy (6.8% vs. 0.5%, p<0.0001).
(c) Intensive insulin therapy was not associated with a benefit in ICU or hospital length of stay, days of mechanical ventilation, or need for renal replacement therapy.
v. A 2023 randomized study of 9230 medical and surgical ICU patients compared liberal glucose control (insulin initiated when BG greater than 215 mg/dL) to intensive insulin control (BG goal 80–110 mg/dL).
(a) Median absolute difference in daily BG between groups was about 28 mg/dL.
(b) There were no differences between groups for ICU length of stay, 90-day mortality, or severe hypoglycemia between groups.

e. A meta-analysis that included seven randomized controlled trials and 11,425 patients found that intensive insulin therapy was not associated with a difference in 28-day mortality (OR 1.04; 95% CI, 0.93–1.17) but was associated with a significant increase in the incidence of hypoglycemia (OR 7.7; 95% CI, 6.0–9.9).

f. The American Diabetes Association and the 2021 Surviving Sepsis Guidelines recommend initiating insulin for critically ill patients with BG exceeds 180 mg/dL.
i. A target of 140–180 mg/dL is recommended for most patients.
ii. A more stringent goal of 110–140 mg/dL could be considered for some patients if it can be achieved without inducing hypoglycemia.

D. Hypoglycemia

1. Clinical presentation

a. Glucose is an obligate molecule for brain function. The brain cannot produce glucose, efficiently use alternative fuels, or store substantial amounts of glycogen, so maintenance of brain function requires a continuous supply of glucose from the circulation.

b. Multiple mechanisms exist to maintain blood glucose within physiologic range. These complex pathways are multifaceted and include pancreatic (decreased insulin and increased glucagon), central nervous system (CNS) (increased norepinephrine and acetylcholine), adrenal medulla (increased epinephrine), and liver (increased glycogenolysis and gluconeogenesis) involvement.

c. In the early stages of hypoglycemia, symptoms such as sweating, anxiety, hunger, palpitations, tremor, and arousal are present. As hypoglycemia persists and worsens, confusion, dizziness, and difficulty speaking develop. Severe hypoglycemia leads to seizures and hypoglycemic coma.

d. Typically, complete recovery of symptoms occurs with glucose administration, but permanent brain damage may occur in patients with severe prolonged hypoglycemia.

e. Hypoglycemia, a reversible cause of cardiac arrest, should be considered for patients with an unclear reason for cardiac arrest.

f. Neurologic symptoms of hypoglycemia may be masked by sedation, and the counterregulatory response may be impaired or masked (e.g., in circulatory shock).

2. Causes

a. Hypoglycemia can be caused by excessive insulin administration, reduced intake of glucose (rarely the cause of severe hypoglycemia in the absence of insulin administration), decreased insulin resistance (weight loss, adrenal or pituitary insufficiency), decreased clearance of insulin (renal or hepatic insufficiency), or other drugs (commonly sulfonylureas, meglitinides, and ethanol; possibly pentamidine, quinine, quinolones, insulin-like growth factor 1, β-blockers, or ACE inhibitors).

b. In a retrospective cohort study of mixed ICU patients, independent predictors of hypoglycemia (BG less than 45 mg/dL) included continuous venovenous hemofiltration with bicarbonate-based replacement solution (OR 14; 95% CI, 1.8–106), a decrease of nutrition without adjustment for insulin infusion (OR 6.6; 95% CI, 1.9–23), diabetes (OR 2.6; 95% CI, 1.5–4.7), insulin use (OR 5.3; 95% CI, 2.8–11), sepsis (OR 2.2; 95% CI, 1.2–4.1), and inotropic support (OR 1.8; 95% CI, 1.1–2.9).
c. Other studies of critically ill patients have identified renal insufficiency, diabetes, mechanical ventilation, female sex, greater severity of illness, longer ICU stay, liver disease, immunocompromised state, and medical or nonelective admission as risk factors for hypoglycemia.
d. Renal insufficiency in the setting of insulin administration should be particularly noted because patients with renal failure have decreased clearance of insulin, which may prolong the duration of hypoglycemia.

3. Management
   a. For conscious patients without severe symptoms, oral glucose should be ingested (juice, soda, or dextrose tablets).
      i. The typical initial glucose dose is 15 g, which should be repeated in 15–20 minutes if symptoms have not improved or if BG remains low.
      ii. Because the response to this "rescue glucose" is transient (less than 2 hours), it should be followed by more substantial glucose intake as a snack or meal.
   b. Parenteral therapy is necessary for patients unable to take glucose orally or hospitalized patients with moderate or severe hypoglycemia.
      i. Glucagon 1 mg promotes hepatic glucogenesis and glycogenolysis. It is a useful therapy for patients with type 1 diabetes (often outside a health care setting) and those without diabetes, but it is less useful in patients with type 2 diabetes because it stimulates insulin secretion and glycogenolysis. Nausea/vomiting is a common adverse effect of glucagon.
      ii. Hospitalized patients with severe hypoglycemia (either severe symptoms or BG less than 40 mg/dL) or those receiving an insulin infusion with BG less than 70 mg/dL (less than 100 mg/dL in neurologic injured patients) should be treated with intravenous dextrose.
         (a) Insulin infusion should be discontinued, as appropriate.
         (b) Administer 10–20 g of 50% dextrose solution (20–40 mL of 50% dextrose in water).
         (c) The BG measurement should be repeated in 15 minutes, with additional dextrose doses administered to achieve a BG greater than 70 mg/dL.
         (d) Care should be taken to avoid excessive dextrose administration in order to avoid iatrogenic hyperglycemia and because glucose variability has been associated with adverse outcomes.
         (e) To prevent hypoglycemia, the insulin regimen should be reassessed if BG is less than 100 mg/dL.

E. Thyroid Crisis
   1. Clinical presentation
      a. Thyroid crisis is characterized by severe manifestations of hyperthyroidism.
      b. Diagnosis can be established on the basis of clinical presentation in a patient with laboratory evidence of hyperthyroidism.
         i. Hyperthermia and diaphoresis are cardinal manifestations of thyroid crisis. Other key components include tachycardia out of proportion to fever and gastrointestinal dysfunction (nausea, vomiting, diarrhea, jaundice).
         ii. In progressive thyroid crisis, patients may develop mental status changes (e.g., agitation, psychosis, coma), and altered mentation has been associated with higher mortality. Other clinical manifestations include cardiac arrhythmias, heart failure, and tachypnea.

iii. Serum TSH is often below the normal range with elevated free T_3 and free and total T_4. Total T_3 levels may be within normal limits in patients with a precipitating illness that reduces T_4 to T_3 conversion.
iv. Other laboratory abnormalities include hypercalcemia, hyperglycemia, leukocytosis or leukopenia, and elevated liver function tests.
v. No universally accepted diagnostic criteria exist, but the Burch and Wartofsky criteria (Table 7) can be used to identify potential thyroid storm. Although this scoring system is effective, it is not highly specific.

Table 7. Burch and Wartofsky Criteria for Thyroid Storm

Criteria	Points
Temperature (°F)	
• 99–99.9	5
• 100–100.9	10
• 101–101.9	15
• 102–102.9	20
• 103–103.9	25
• ≥ 104	30
Heart rate (beats/minute)	
• 99–109	5
• 110–119	10
• 120–129	15
• 130–139	20
• ≥ 140	25
Atrial fibrillation	
• Absent	0
• Present	10
Congestive heart failure	
• Absent	0
• Mild (edema)	5
• Moderate (rales)	10
• Severe (pulmonary edema)	20
GI signs	
• Absent	0
• Moderate (nausea, vomiting, diarrhea, abdominal pain)	10
• Severe (jaundice)	20
CNS disturbance	
• Absent	0
• Mild (agitation)	10
• Moderate (delirium, psychosis, extreme lethargy)	20
• Severe (seizure, coma)	30
Precipitant history	
• Positive	0
• Negative	10

Table 7. Burch and Wartofsky Criteria for Thyroid Storm *(continued)*

Criteria	Points
Score • > 44: Thyroid storm • 25–44: Suggestive/impending storm • < 25: Storm unlikely	

Information from: Bahn RS, Burch HB, Cooper DS, et al. Hyperthyroidism and other causes of thyrotoxicosis: management guidelines of the American Thyroid Association and American Association of Clinical Endocrinologists. Thyroid 2011;21:593-646.

2. Causes
    a. Thyroid crisis can develop in patients with a history of untreated hyperthyroidism but is more often precipitated by an acute event. Precipitating events include surgery, pregnancy, and trauma. In hospitalized patients, the most common cause is infection.
    b. Drugs associated with thyroid crisis include radioactive iodine, overdose of levothyroxine or liothyronine, cytotoxic chemotherapy, aspirin, iodinated contrast dye, amiodarone, and organophosphate toxicity.
    c. Rarely, thyroid crisis is the initial presentation for patients with undiagnosed hyperthyroidism.
3. Management
    a. Treatment of thyroid crisis is complex and requires several interventions simultaneously.
    b. Decrease thyroid hormone synthesis.
        i. Methimazole and propylthiouracil inhibit production of new thyroid hormone.
        ii. Propylthiouracil is preferred in thyroid crisis because it not only blocks new thyroid synthesis but also decreases peripheral conversion of T_4 to T_3.
        iii. Propylthiouracil is also preferred in the first trimester of pregnancy; after the start of the second trimester, patients may continue propylthiouracil or transition to methimazole. Concentrations of both agents in breast milk are low, but methimazole is generally preferred in breastfeeding mothers due to risk of hepatotoxicity with propylthiouracil.
        iv. Propylthiouracil is given as a 500- to 1000-mg loading dose, followed by 250 mg every 6 hours. Doses may be given enterally or rectally.
        v. Patients receiving propylthiouracil should be monitored for adverse effects, including agranulocytosis, allergic hepatitis, and vasculitis.
        vi. If methimazole is used because of propylthiouracil allergy or intolerance, usual dose is 20–30 mg every 8 hours. Doses may be given enterally or rectally.
        vii. Intravenous steroids (hydrocortisone 300-mg loading dose, followed by 100 mg every 8 hours) should be initiated to block the conversion of T_4 to T_3.
        viii. Plasmapheresis has been used to remove cytokines, antibodies, and thyroid hormones from plasma when traditional therapies have not been successful. Effects are transient, lasting 24–48 hours, but improvement in clinical status has enabled patients to be stabilized for definitive therapy with thyroidectomy.
    c. Inhibit thyroid hormone release with inorganic iodine.
        i. Propylthiouracil and methimazole inhibit thyroid hormone synthesis but do not block release of stored hormone from the thyroid gland.
        ii. Initial dose of iodine should be given at least 1 hour after the first dose of propylthiouracil or methimazole to reduce the risk of providing more substrate for thyroid hormone production.
        iii. Oral potassium iodide (Lugol solution) is administered as 3–5 drops every 6 hours. If available, sodium iodide can be given as an intravenous infusion of 0.5–1 g over 24 hours.
        iv. In patients who are allergic to iodine, lithium carbonate may be used as an alternative. Initial dose should be 300 mg every 6 hours, with subsequent adjustment to maintain serum lithium concentration of 0.8–1.2 mEq/L.

d. Reduce levels of circulating thyroid hormone.
   i. Cholestyramine can be given to bind circulating T_3 and T_4.
   ii. Salicylates should be avoided because use can decrease binding of thyroid hormones to proteins and therefore increase concentrations of free thyroid hormones.
   iii. In severe cases that do not respond to standard therapy, plasmapheresis and therapeutic plasma exchange may be considered to decrease T_4 and T_3 levels.
e. Reduce heart rate.
   i. Because cardiovascular collapse leads to systemic decompensation, β-blockers should be initiated as quickly as possible.
   ii. β-Blocker dose should be titrated to achieve heart rate control (typically below 90 beats/minute).
   iii. Traditionally, propranolol is the preferred therapy because it may block T_4 to T_3 conversion at high doses. Propranolol is also the agent of choice in women who are pregnant or breastfeeding. Dosing for this indication is aggressive, starting at 60 mg orally every 4 hours with an optional initial loading dose of 80 mg.
   iv. Alternative agents include carvedilol, esmolol, and diltiazem. Diltiazem should be reserved for patients with active bronchospasm or for those who do not tolerate β-blockers.

F. Myxedema Coma
1. Clinical presentation
   a. Myxedema coma occurs in patients with advanced untreated hypothyroidism. No absolute diagnostic criteria distinguish myxedema coma from severe hypothyroidism. Identifying telltale signs of hypothyroidism is key to making the appropriate diagnosis in a timely manner.
   b. Diagnosis is made on the basis of clinical presentation, not laboratory evidence of hypothyroidism.
      i. Altered mental status and hypothermia are the most common presenting symptoms. Other clinical features include hypotension, bradycardia, hypoglycemia, constipation, urinary retention, puffiness of the hands and face, and pleural, pericardial, or peritoneal effusions.
      ii. Patients may present with shock or arrhythmias, including prolongation of the QT interval that can lead to torsades de pointes.
      iii. If a patient presents with signs of infection without the systemic inflammatory response syndrome, myxedema coma should be suspected.
      iv. Patients typically have elevated TSH and low or undetectable T_3 and T_4, but TSH may be inappropriately low or normal in cases of central hypothyroidism. The level of change in thyroid hormones does not correlate well with disease severity.
   c. Myxedema coma is more common in older women and occurs more often during winter months because of altered temperature regulation.
   d. Discussion of nonthyroidal illness syndrome ("euthyroid sick syndrome" or "SICU thyroid") is beyond the scope of this chapter.
2. Causes
   a. Myxedema coma may be the consequence of longstanding hypothyroidism but can also be precipitated by an acute event such as infection, myocardial infarction, surgery, or cold exposure.
   b. Drugs associated with the development of myxedema coma include sedatives, anesthetics, narcotics, lithium, immune checkpoint inhibitors, and amiodarone, as well as abrupt discontinuation of levothyroxine.
3. Management
   a. Because myxedema coma is an endocrine emergency, treatment should be initiated without waiting for laboratory data.
   b. General treatment measures include rewarming of the patient and treatment of the precipitating illness. Because infection is a common precipitator of myxedema coma, all patients should undergo a thorough infectious workup.

c. Guidelines from the American Thyroid Association recommend the following for patients with myxedema coma:
   i. Patients may have underlying adrenocorticotropic hormone deficiency, and restoration of thyroid function can accelerate cortisol metabolism. Empiric intravenous corticosteroids at doses appropriate for the stressed state should be administered before levothyroxine (strong recommendation). Hydrocortisone is typically given in doses of 50 mg every 6 hours or 100 mg every 8 hours.
   ii. Thyroid hormone replacement should be initiated with intravenous levothyroxine. A loading dose of 200–400 mcg should be given, with lower doses used in patients who are smaller or older or who have a history of cardiac disease or arrhythmia (strong recommendation).
   iii. After the initial loading dose, the daily replacement dose should be 1.6 mcg/kg body weight intravenously (strong recommendation). Daily therapy may be changed to the enteral route after the patient improves clinically.
   iv. Because T_4 to T_3 conversion may be decreased in myxedema coma, intravenous liothyronine may be given in addition to levothyroxine (weak recommendation). High serum T_3 during treatment is associated with mortality, so high doses should be avoided. A loading dose of 5–20 mcg can be given, followed by 2.5–10 mcg every 8 hours. Patients who are smaller or older or who have a history of cardiac disease or arrhythmia should receive smaller doses. Treatment can be continued until the patient has regained consciousness and clinical parameters have improved.
   v. Patients should be monitored for improvements in mental status, cardiac function, and pulmonary function.
   vi. Measurement of thyroid hormones every 1–2 days is reasonable to ensure a favorable trajectory. Optimal levels of TSH, T_4, and T_3 in this scenario have not been defined, but lack of improvement may be an indication to increase the levothyroxine dose and/or add liothyronine. High serum T_3 can be considered an indication to decrease therapy for safety reasons (weak recommendation).

d. Patients may need supportive care with vasoactive medications. Typically, a catecholamine agent with β_1-adrenergic receptor agonist properties (epinephrine or dopamine) is preferred to increase the patient's heart rate and blood pressure.

Patient Case

8. A 23-year-old man (weight 80 kg) presents to the ED with an acute mental status change and a core body temperature of 94°F (34.4°C). He has a history of hypothyroidism, but according to his family, he had decided to stop taking all of his thyroid medications 1 week earlier. The team has given him a diagnosis of myxedema coma. Which is the most appropriate treatment option for this patient?
   A. Intravenous levothyroxine 400 mcg × 1 followed by 125 mcg daily.
   B. An insulin infusion titrated to a BG of 140–180 mg/dL.
   C. Propylthiouracil 200 mg every 4 hours.
   D. Propranolol 60 mg orally every 4 hours.

G. Adrenal insufficiency
   1. Clinical presentation
      a. Adrenal crisis
         i. Because there is no universally accepted evidence-based definition of adrenal crisis, cases are identified based on a pragmatic definition and clinical assessment. In adults, adrenal crisis is defined as an acute deterioration in health status associated with absolute or relative hypotension (SBP less than 100 mm Hg or 20 mm Hg or more below baseline) that improves within 1–2 hours after parenteral glucocorticoid administration. Adrenal crisis is a rare cause of vasodilatory shock.
         ii. Patients with acute adrenal insufficiency typically present with severe hypotension, altered mental status, acute abdominal pain, vomiting, and fever. As such, patients often receive a misdiagnosis of an acute abdomen. Accompanying symptoms may include fatigue, lethargy, confusion, weakness, and pain in the muscles or joints. Patients with primary adrenal insufficiency may also have salt craving (because of mineralocorticoid deficiency secondary to a lack of aldosterone). Characteristic skin hyperpigmentation from primary adrenal insufficiency may also be present.
         iii. Laboratory findings will include low serum cortisol (i.e., cortisol level below 18 mcg/dL in an acutely stressed patient is suggestive of adrenal insufficiency); hyponatremia, hyperkalemia, hypoglycemia, and normocytic anemia may also be present.
      b. Critical illness-related corticosteroid insufficiency (CIRCI)
         i. CIRCI refers to impairment of the hypothalamic-pituitary-adrenal axis during critical illness as a complication of an underlying disease. Patients most commonly have septic shock but may also have acute respiratory distress syndrome, cardiac arrest, trauma, or burns.
         ii. Patients with CIRCI often have hypotension refractory to fluids that requires addition of vasopressors. Laboratory assessment may show hypoglycemia and eosinophilia; hyponatremia and hyperkalemia are uncommon.
         iii. Historically, CIRCI was diagnosed according to either a low serum cortisol concentration or an inadequate increase in cortisol after a cosyntropin stimulation test. These tests have significant limitations in critically ill patients, such as poor reproducibility of the cosyntropin stimulation test and lack of agreement of commercially available cortisol assays with analytic standards.
         iv. Despite these limitations, an increase in cortisol after a cosyntropin stimulation test less than 9 mg/dL is the best predictor of adrenal insufficiency (as determined by metyrapone testing in patients with severe sepsis/septic shock). A serum cortisol value less than 10 mg/dL has a high positive predictive value for adrenal insufficiency but a low sensitivity.
         v. Corticosteroids should only be given for patients with septic shock and suspected CIRCI who do not achieve resuscitation goals despite fluid administration and a ongoing vasopressor requirement.
         vi. Of importance, clinical practice guidelines for the treatment of CIRCI recommend against using cortisol-based testing in determining a patient's candidacy for corticosteroid therapy. As such, using cortisol assays or a cosyntropin stimulation test to diagnose CIRCI and influence treatment decisions is not recommended.

2. Causes
    a. Adrenal crisis may result from either primary adrenal failure or secondary adrenal disease because of impairment of the hypothalamic-pituitary axis.
    b. Acute-onset primary adrenal insufficiency is usually caused by adrenal hemorrhage or infarction, whereas acute-onset secondary adrenal insufficiency is usually caused by pituitary apoplexy, pituitary or hypothalamic surgery, or traumatic brain injury. Patients receiving corticosteroids who have acute stress (e.g., infection or surgery) will be unable to increase their cortisol appropriately because of suppression of corticotropin-releasing hormone and corticotropin.
    c. Bacterial infections commonly precipitate adrenal crisis in adults. In children, viral infections are more common precipitating events.
    d. Several medications may contribute to inadequate serum cortisol concentrations, including etomidate and ketoconazole. Adrenal crisis has been reported after vaccinations and zoledronic acid infusions. Immune checkpoint inhibitor therapy may also cause adrenal insufficiency.
    e. An abrupt decrease in dose or cessation of corticosteroids in patients receiving moderate to high glucocorticoid doses (equivalent to prednisone 7.5 mg daily or above) for 3 weeks or more may precipitate adrenal crisis.
    f. CIRCI is caused by inadequate circulating cortisol concentrations combined with tissue resistance.
3. Management
    a. Acute adrenal crisis
        i. Patients with acute adrenal crisis should receive intravenous fluids and vasopressors (typically isotonic sodium chloride and norepinephrine) for the treatment of shock (if present). After initial fluid bolus, dextrose may be added for treatment of hypoglycemia.
        ii. Corticosteroids should be administered, but the optimal dose is unclear. Typically, intravenous hydrocortisone at 200–300 mg/day (50 mg every 6 hours or 100 mg every 8 hours) is initiated. Some experts have advocated that lower doses are sufficient.
        iii. Although hydrocortisone is preferred because of its protein binding, tissue distribution, and balanced glucocorticoid-mineralocorticoid effects, other parenteral glucocorticoids may be used. Dosing is based on glucocorticoid potency relative to hydrocortisone. Mineralocorticoid replacement is usually unnecessary in the acute setting because its effect on sodium retention takes several days to develop.
        iv. After the patient is clearly improved (e.g., shock resolution), the corticosteroid dose can be tapered and may be transitioned to oral therapy.
    b. Patients receiving chronic corticosteroids who have an acute stressor should receive an increased glucocorticoid dose ("stress dose") of corticosteroids to prevent adrenal crisis.
        i. The corticosteroid dose recommended in this setting depends on the level of stress.
        ii. In febrile patients, those undergoing major dental procedures, and those undergoing invasive diagnostic procedures (e.g., colonoscopy), a doubling or tripling of the maintenance dose for 1–3 days may be sufficient.
        iii. Patients experiencing severe infection, undergoing major surgery, or presenting with shock should have intravenous hydrocortisone at 200–300 mg/day (50 mg every 6 hours or 100 mg every 8 hours) initiated. This dose should be tapered to the patient's home regimen after the patient clinically improves or after surgery.
    c. Treatment of CIRCI in the setting of septic shock is discussed in the Shock Syndromes I: Introduction, Vasodilatory, and Sepsis chapter. Use of corticosteroids for the treatment of acute respiratory distress syndrome is discussed in the Pulmonary Disorders I chapter.

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ANSWERS AND EXPLANATIONS TO PATIENT CASES

1. **Answer: B**

Intravenous acetylcysteine increases transplant-free survival rates for patients with NAI-ALF, particularly for patients with low-grade encephalopathy. Oral acetylcysteine has not been studied in a randomized controlled trial for NAI-ALF (Answer C is incorrect), and the dosing strategy for NAI-ALF is different from the 21-hour intravenous regimen for acetaminophen overdose (Answer A is incorrect). The dosing strategy for NAI-ALF is a 72-hour regimen with a 150-mg/kg bolus, followed by a 12.5-mg/kg/hour dose for 4 hours and then a 6.25-mg/kg/hour dose for 67 hours (Answer B is correct). Oral glutamine is not used for NAI-ALF; it has been studied to aid in the healing of GI fistulas (Answer D is incorrect).

2. **Answer: B**

Osmotic agents are first-line treatment for control of ICP. Although hypertonic saline prevents ICP elevations, the continuous infusion is not used for acute control (Answer A is incorrect). For acute control of ICP elevations, mannitol boluses are used first line provided serum osmolaity is less than 320 (Answer B is correct). Hyperventilation and barbiturates are only used to control ICP elevations when other options have failed (Answers C and D are incorrect).

3. **Answer: C**

This patient has severe acute necrotizing pancreatitis because she has not improved after the first 48 hours, and her CT reveals pancreatitic necrosis involving more than 30% of her pancreas. There appears to be no benefit with using prophylactic antibiotics for patients with necrotizing AP, in reducing either mortality rates or rates of pancreatic and extrapancreatic infections, particularly in more recent studies (Answers A and D are incorrect). Surgical management for sterile necrosis is only recommended if patients have gastric outlet obstruction and/or bile duct obstruction (Answer B is incorrect). For patients with severe acute necrotizing pancreatitis, it is recommended to defer antibiotics unless there is suggestion of infection or if patients have not improved within 7–10 days (Answer C is correct).

4. **Answer: D**

Somatostatin significantly decreases fistula output compared with placebo (Answer A is incorrect). Total parenteral nutrition increases spontaneous closure rates by reducing GI secretions (Answer C is incorrect), and octreotide had a beneficial effect on fistula output in one small study (Answer B is incorrect). Although glutamine has been associated with spontaneous rates of fistula closure, it has been not been associated with reduced fistula output (Answer D is correct).

5. **Answer: B**

Of the possible answers, only alvimopan has been shown to reduce the incidence of ileus postoperatively (Answer B is correct). Metoclopramide has shown mixed results, though the antiemetic properties may be beneficial as adjunctive therapy in POI, and octreotide is not used in the prevention of POI (Answers C and D are incorrect). All opioids can contribute equally to POI (Answer A is incorrect).

6. **Answer: A**

According to the simplified Apfel risk score criteria, this patient has four risk factors for developing PONV (female sex, nonsmoker, history of motion sickness, and perioperative opioids). These risk factors place her at high risk of developing PONV, estimated at greater than 80% (Answers B and D are incorrect). Patients with a high risk of PONV should receive more than two pharmacologic interventions to prevent PONV (Answer A is correct; Answer C is incorrect).

7. **Answer: D**

Frank bloody output from the rectum is more indicative of a lower GI bleed than an UGIB. However, patients with a brisk UGIB may present with bright red blood per rectum. Although a lower GI bleed is more likely, the patient should be initiated on a PPI infusion and undergo an esophagogastroduodenoscopy as soon as possible (Answer D is correct). After the esophagogastroduodenoscopy, the patient should have a colonoscopy. The priority in this patient's case is to evaluate for an UGIB because the finding of "coffee-ground" material on NG lavage suggests that the bleeding source is UGIB (Answer C is incorrect). Although previous guidelines recommended initiating PPIs before endoscopy, the

most recent version does not recommend for or against this practice (Answer A is incorrect). High-dose PPIs should be continued for 48 hours after endoscopy and then followed by twice-daily treatment from days 4–14. (Answer B is incorrect).

8. **Answer: A**

The patient, who presented with signs and symptoms of myxedema coma, was given this diagnosis by the treating team. Of the possible answers listed, only levothyroxine is appropriate for the treatment of myxedema coma (Answer A is correct). Insulin, propylthiouracil, and propranolol play no role in the treatment of myxedema coma (Answers B–D are incorrect).

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: C**
The most recent guidelines from the American Association for the Study of Liver Diseases and the U.S. Acute Liver Failure Study Group define ALF as a coagulopathy, usually an INR of 1.5 or more, with any degree of encephalopathy in patients without pre-existing liver disease (Answer C is correct). Although jaundice, thrombocytopenia, and leukocytosis can occur in patients with ALF, they are not currently defined as hallmark signs of the disease that exist in all patients (Answers A, B, and D are incorrect).

2. **Answer: A**
Administration of acetylcysteine is recommended in all ALF cases when acetaminophen is suspected as a cause, regardless of the acetaminophen concentration (Answer B is incorrect). Although oral and intravenous formulations have efficacy for the treatment of acetaminophen overdose, the intravenous formulation is recommended when patients have greater than grade I encephalopathy or hypotension or when they cannot tolerate oral therapy (Answers C and D are incorrect). Intravenous acetylcysteine is recommended for most patients who present with liver failure and can be extended beyond the 21-hour regimen, especially if therapy was initiated more than 8 hours after ingestion or baseline concentrations were greater than 300 mg/dL (Answer A is correct).

3. **Answer: D**
Patients with AP should not be kept NPO (Answer A is incorrect). Studies that have compared TPN with enteral feeding in AP have shown that enteral feeding is associated with reduced mortality and infectious complications. Thus, enteral feeding is recommended over TPN for AP, if it is tolerated (Answer B is incorrect). Enteral feeding can be given by either the NJ or the NG route for AP, though the NG route may increase the risk of aspiration. Because this patient has several admissions for aspiration and is thus high risk for aspiration, it is safer to use the NJ route over the NG route (Answer C is incorrect, Answer D is correct).

4. **Answer: C**
Fistula output is defined as high if the output is greater than 500 mL/day, moderate if it is 200–500 mL/day, and low if it is less than 200 mL/day. This patient's fistula output has decreased significantly (Answers A and D are incorrect) from 570 mL/day to 250 mL/day, but it is still not enough to classify her fistula output as low (Answer B is incorrect). Her output has decreased from high to moderate (Answer C is correct).

5. **Answer: B**
In a long-term study of alvimopan for opioid-induced bowel dysfunction, alvimopan was associated with higher rates of myocardial infarction than was placebo. To mitigate this risk, the FDA has limited the use of alvimopan to short-term, inpatient use, and patients cannot receive more than 15 doses (Answer B is correct); however, its use is not contraindicated in patients with a history of myocardial infarction (Answer D is incorrect). The FDA-approved dose is 12 mg twice daily, and there is no requirement for QTc monitoring with alvimopan (Answers A and C are incorrect).

6. **Answer: D**
Dexamethasone and aprepitant should be given before inducing anesthesia for the prevention of PONV (Answers A and C are incorrect). Ondansetron and other serotonin-3 antagonists are most effective if given at the end of surgery (Answer B is incorrect). Droperidol is effective for the prevention of PONV when given at the end of surgery (Answer D is correct).

7. **Answer: A**
H. pylori is a recognized carcinogen and should be eradicated using a 14-day PPI/antibiotic combination (Answer B is incorrect). Patients with an acute UGIB who present with a high-risk bleed should have a diagnostic and therapeutic endoscopy within 24 hours of admission (Answer C is incorrect). Blood transfusions should be administered to keep the hemoglobin concentration greater than 7 g/dL (Answer D is incorrect). Therefore, the only inappropriate treatment option is using octreotide (Answer A is correct).

8. **Answer: B**

Thyroid storm is an uncommon but deadly manifestation of hyperthyroidism; therefore, TSH will be low, whereas T_4 and T_3 will be high (Answer B is correct). Myxedema coma is a manifestation of hypothyroidism; therefore, patients will typically have high TSH and low T_4/T_3. When TSH is high, both T_3 and T_4 are typically low (Answer A is incorrect). Conversely, if TSH is low, both T_3 and T_4 are typically high (Answers C and D are incorrect).

Appendix 1. Most Common or Well-Described Drugs Associated with DILI and the Patterns of Liver Injury

Drug Class	Typical Pattern of Injury
Antibiotics	
• Amoxicillin/clavulanate	• Cholestatic injury
• Isoniazid	• Hepatocellular injury
• Trimethoprim/sulfamethoxazole	• Cholestatic injury, can also be hepatocellular injury
• Fluoroquinolones	• Hepatocellular, cholestatic, or mixed injury
• Macrolides	• Hepatocellular injury
• Nitrofurantoin	• Hepatocellular injury
• Minocycline	• Hepatocellular injury
Antiepileptics	
• Phenytoin	• Hepatocellular, cholestatic, or mixed injury
• Carbamazepine	• Hepatocellular, cholestatic, or mixed injury
• Lamotrigine	• Hepatocellular injury
• Valproate	• Hepatocellular injury, may also cause hyperammonemia
Nonsteroidal anti-inflammatory drugs	• Hepatocellular
Immune modulators	
• Interferon	• Hepatocellular injury
• Anti-TNF agents	• Hepatocellular injury
• Azathioprine	• Cholestatic injury, can also be hepatocellular injury
Herbal medications and dietary supplements	
• Green tea extract	• Hepatocellular injury
• Flavocoxid	• Hepatocellular, cholestatic, or mixed injury
Miscellaneous	
• Methotrexate	• Fatty liver
• Allopurinol	• Hepatocellular or mixed injury
• Amiodarone	• Hepatocellular, cholestatic, or mixed injury
• Androgen-containing steroids	• Cholestatic injury
• Inhaled anesthetics	• Hepatocellular injury
• Niacin (more common with long-acting form)	• Hepatocellular injury
• Sulfasalazine	• Hepatocellular, cholestatic, or mixed injury
• Proton pump inhibitors	• Hepatocellular injury

TNF = tumor necrosis factor.

Information from: Chalasani NP, Hayashi PH, Bonkovsky HL, et al. ACG clinical guideline: the diagnosis and management of idiosyncratic drug-induced liver injury. Am J Gastroenterol 2014;109:950-66.